

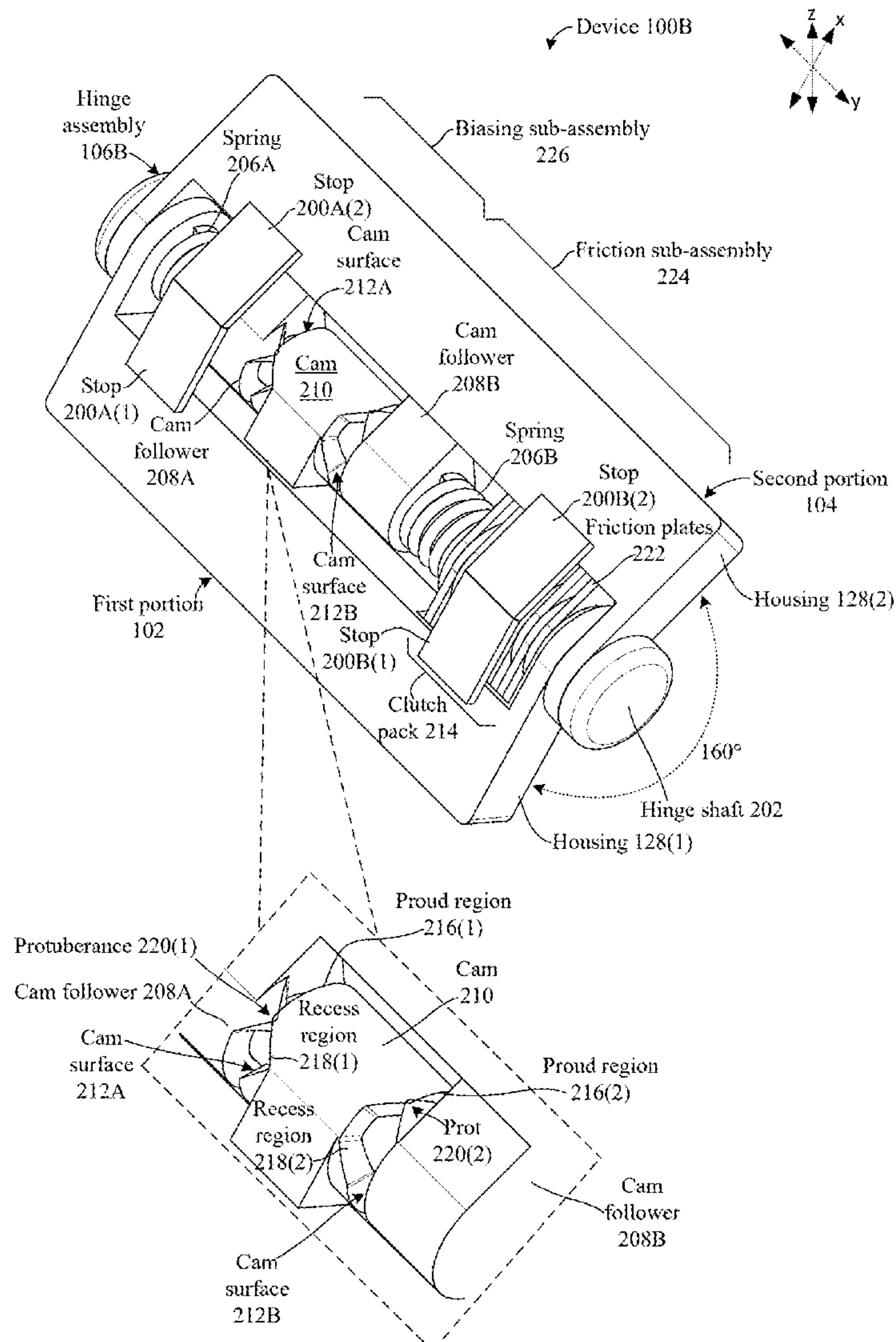
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(19) **United States**(12) **Patent Application Publication**
CHONG et al.(10) **Pub. No.: US 2026/0143604 A1**(43) **Pub. Date: May 21, 2026**(54) **MULTI-FUNCTION HINGE**(71) Applicant: **Microsoft Technology Licensing, LLC**,
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(2013.01)

(57)

ABSTRACT

The description relates to hinged devices, such as hinged computing devices and accessories. One example can include a first portion and a second portion that are rotatably secured by a hinge shaft through a range of rotation from a closed orientation to a fully open orientation. A friction sub-assembly can be positioned on the hinge shaft and configured to create resistance to rotation during a first sub-range of rotation that ranges from the fully opened orientation to an intermediary orientation. A biasing sub-assembly can be positioned on the hinge shaft and configured to create a bias force during a second sub-range of rotation that ranges from the intermediary orientation to the closed orientation and the biasing force biases the first and second portions to an individual orientation within the second sub-range of rotation.



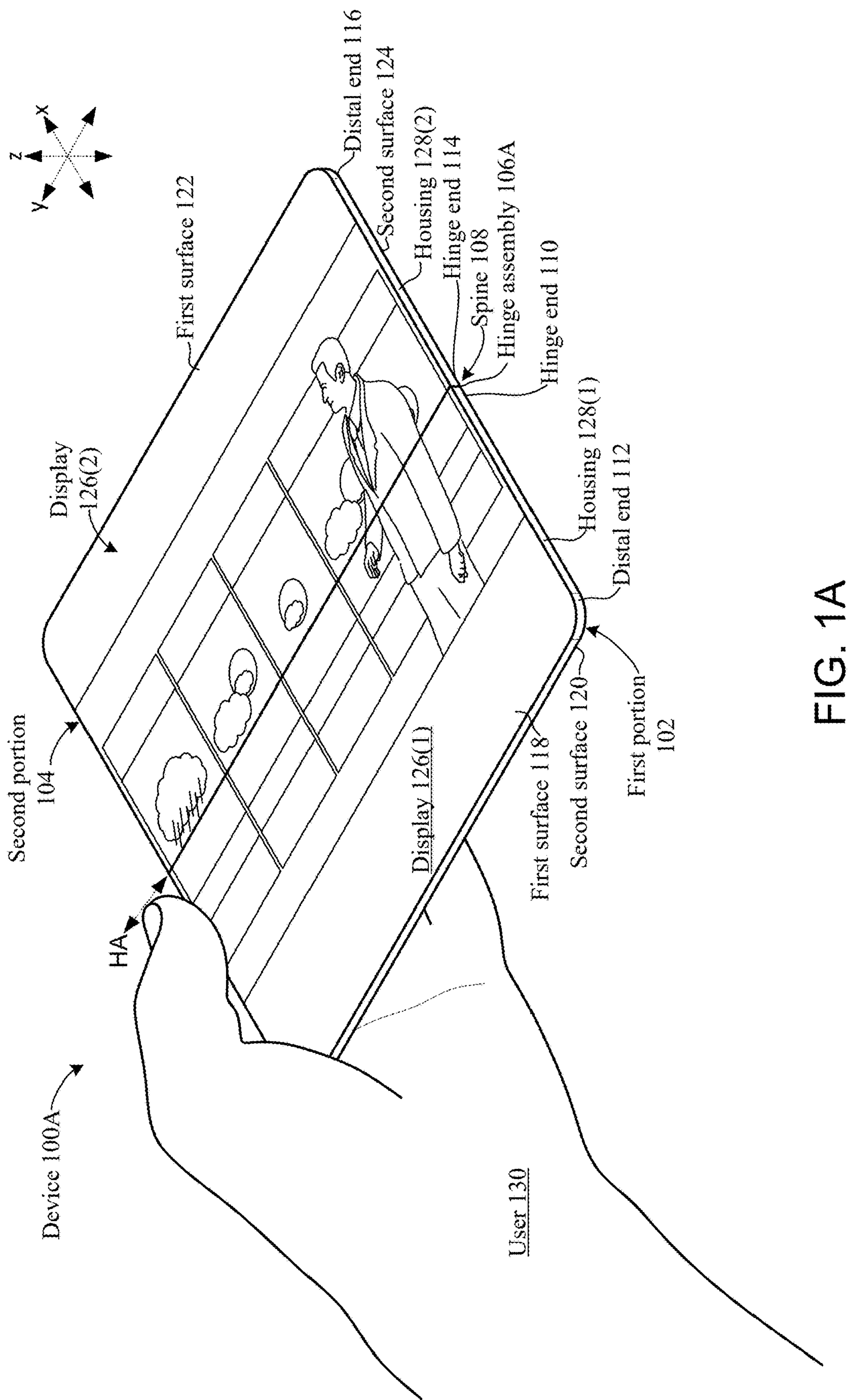


FIG. 1A

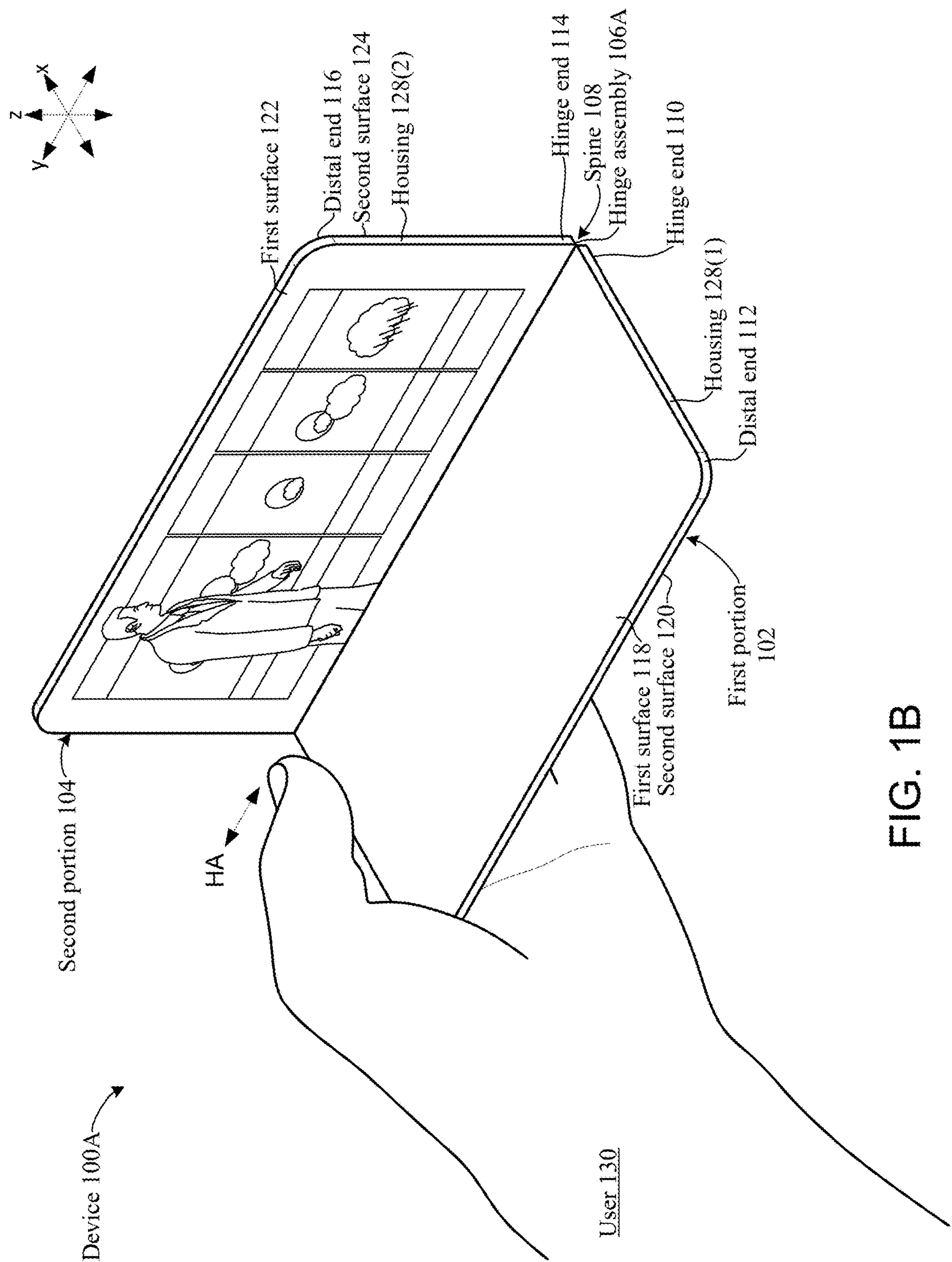


FIG. 1B

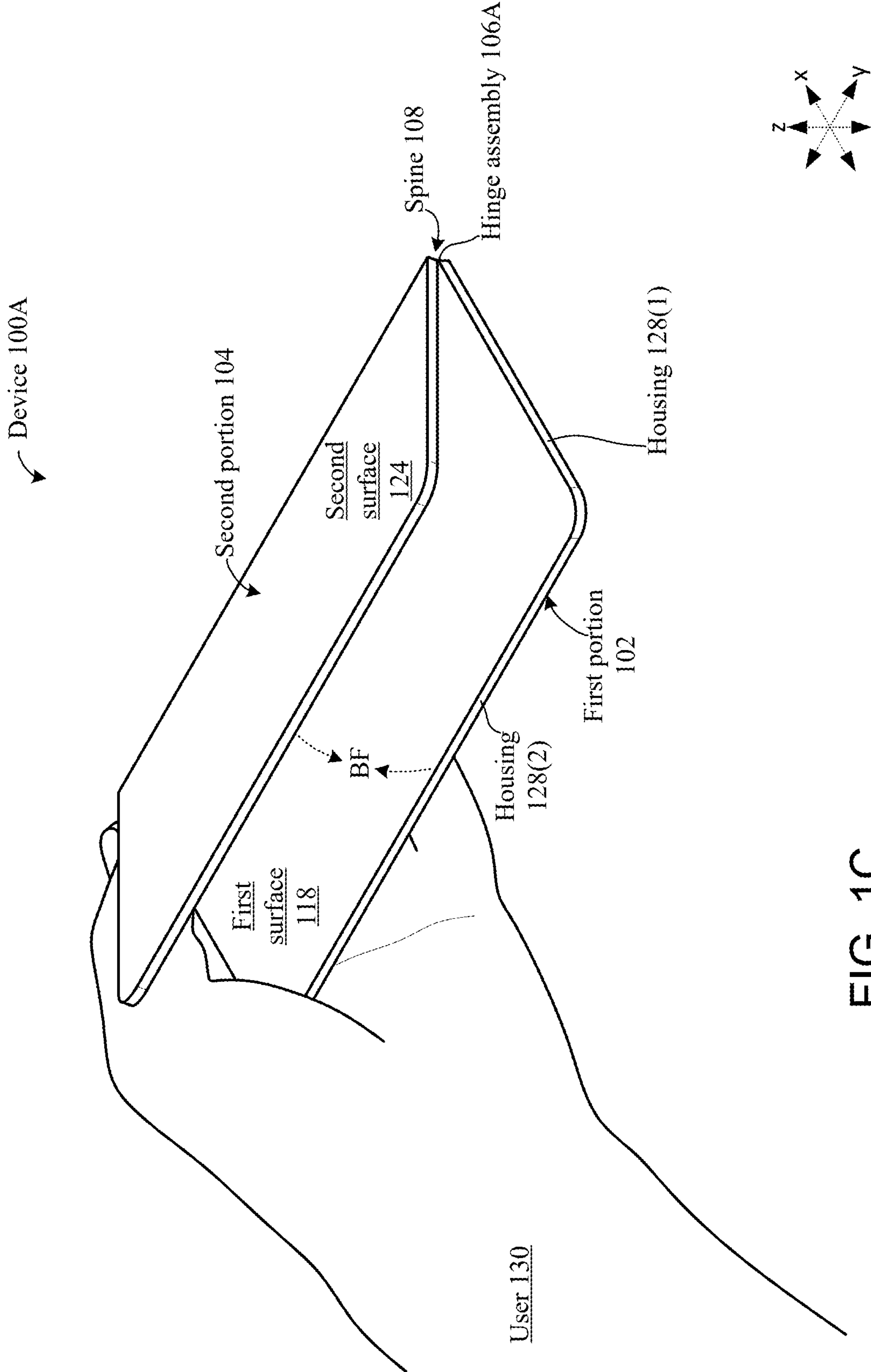


FIG. 1C

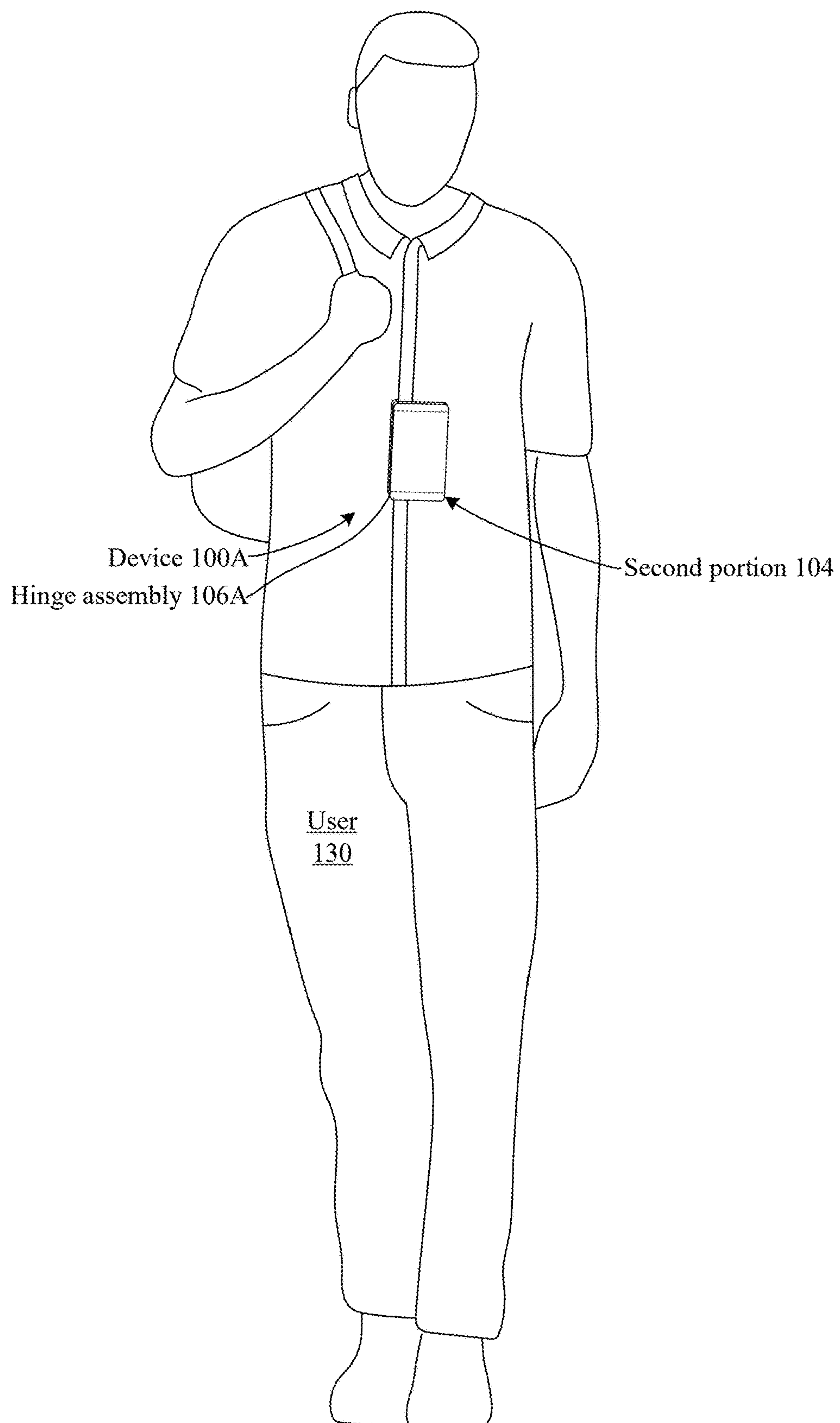


FIG. 1D

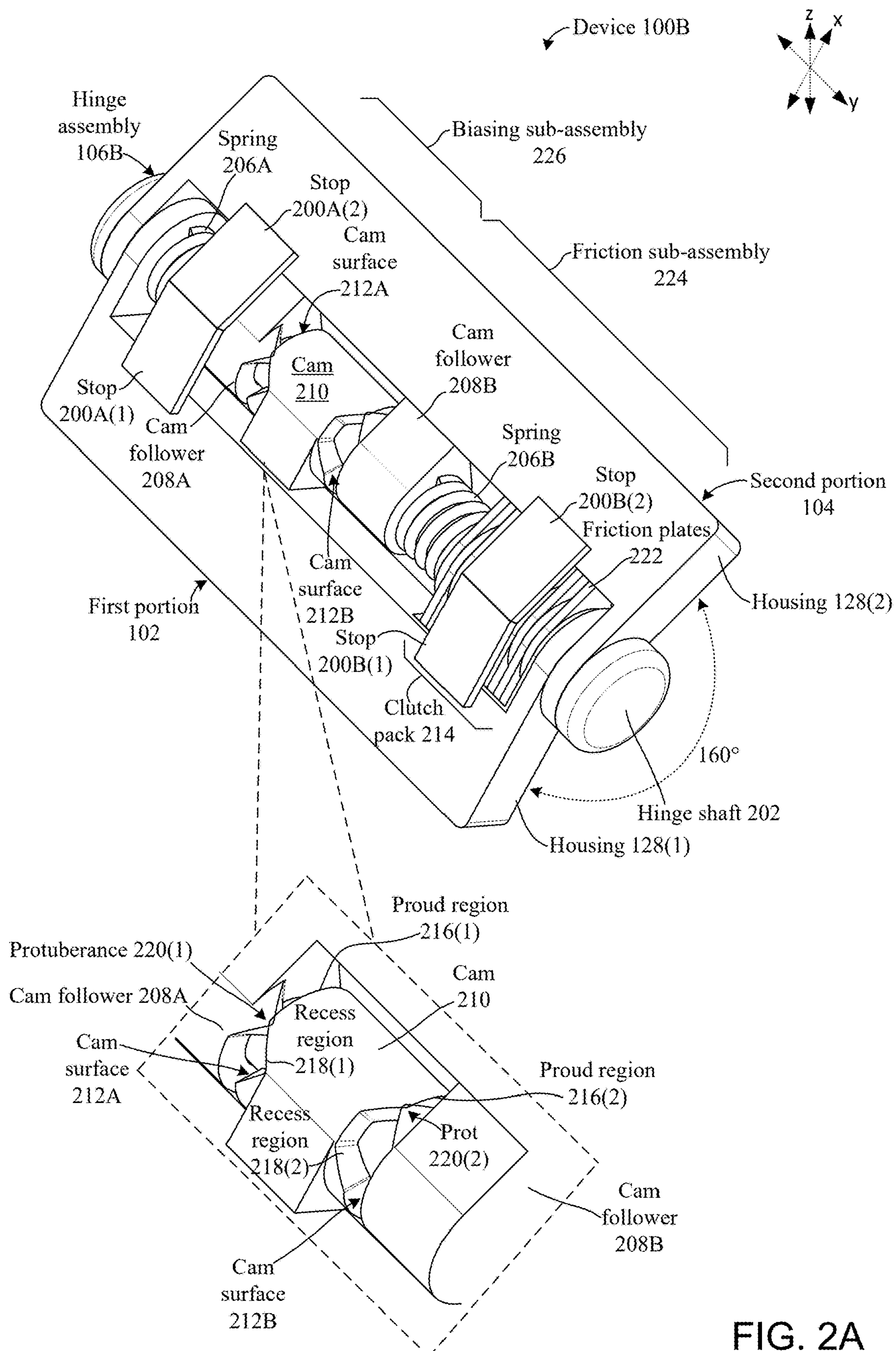


FIG. 2A

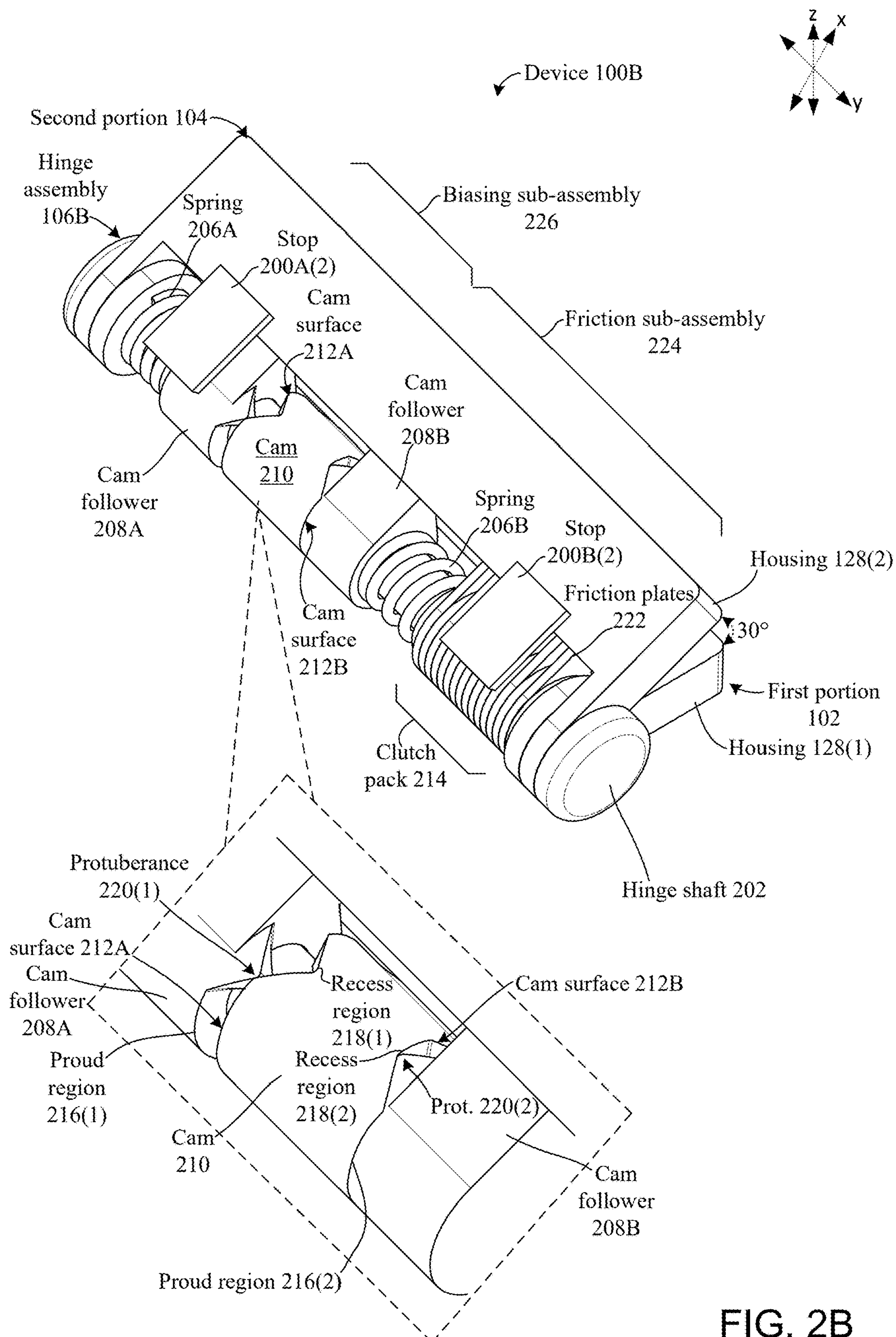


FIG. 2B

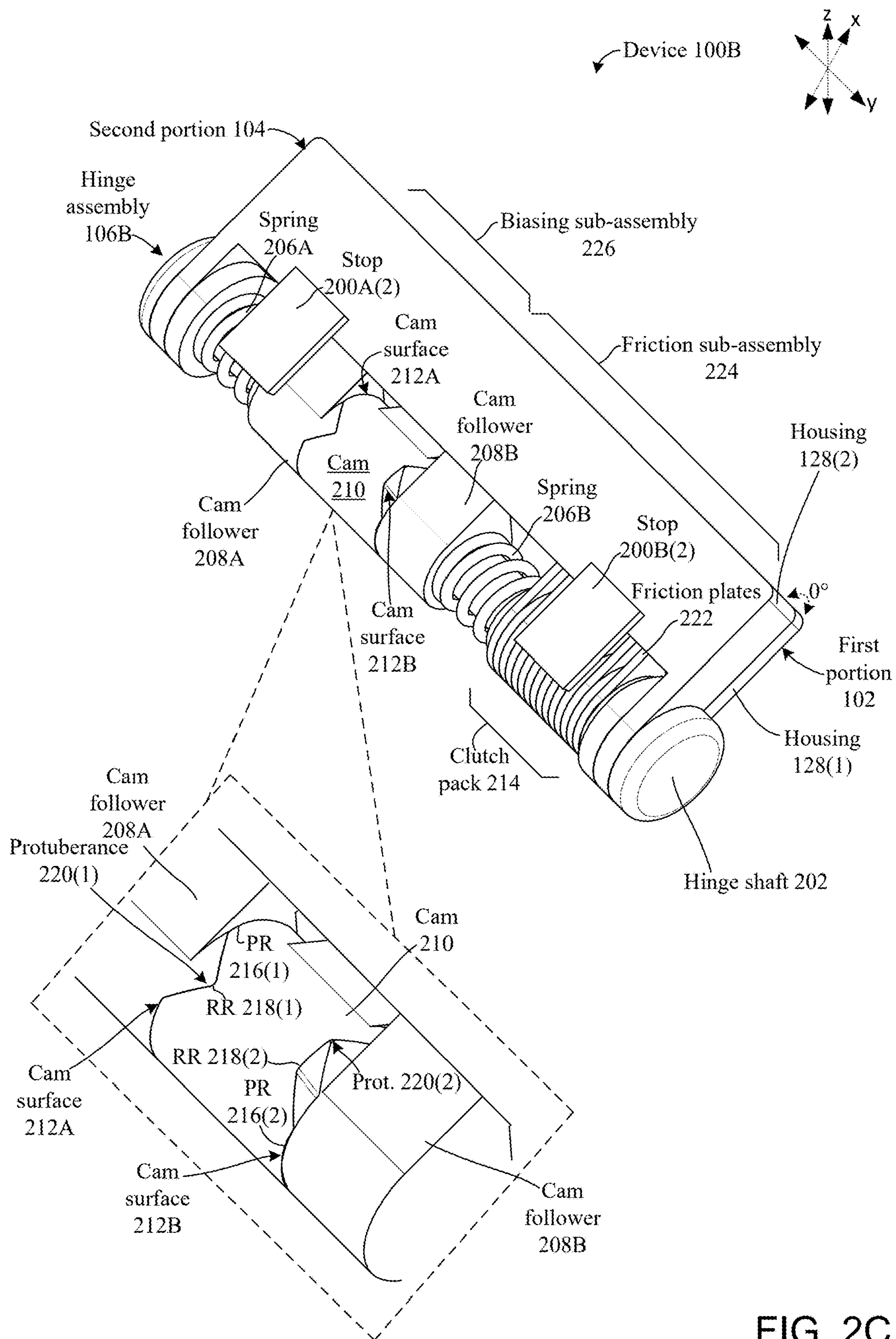


FIG. 2C

Device 100C

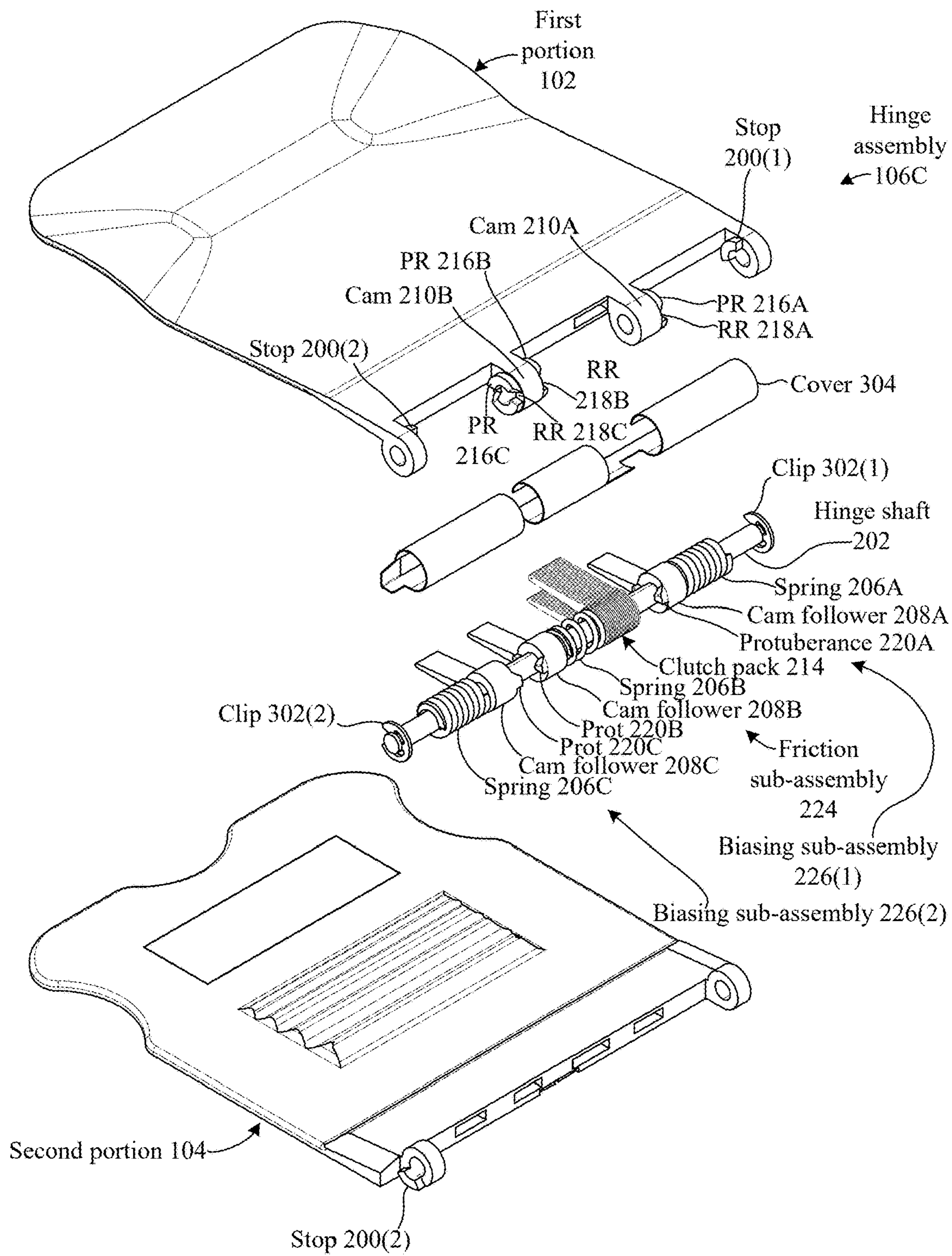
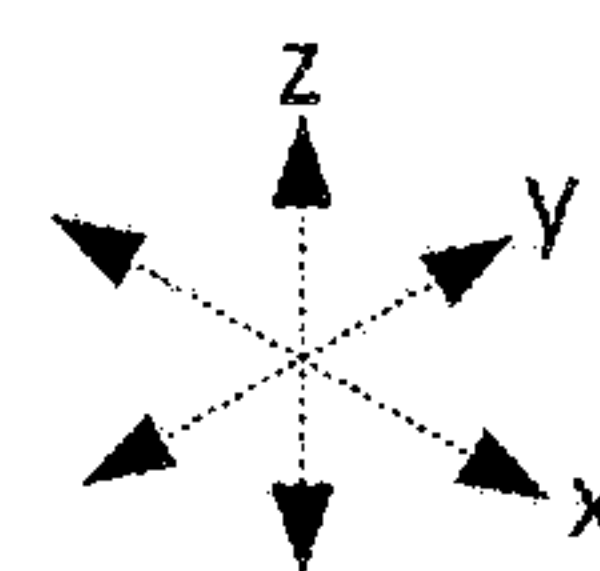


FIG. 3A

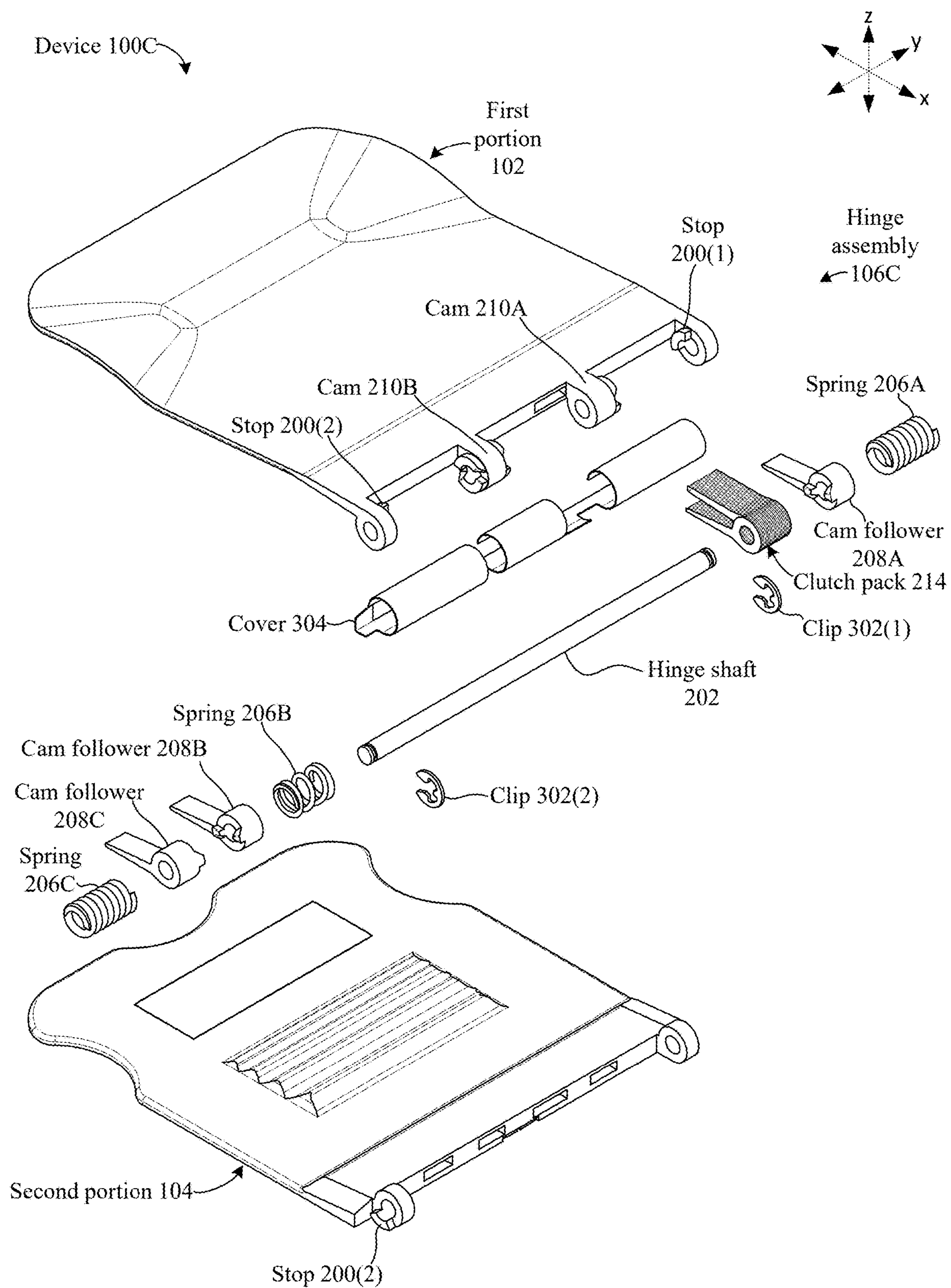


FIG. 3B

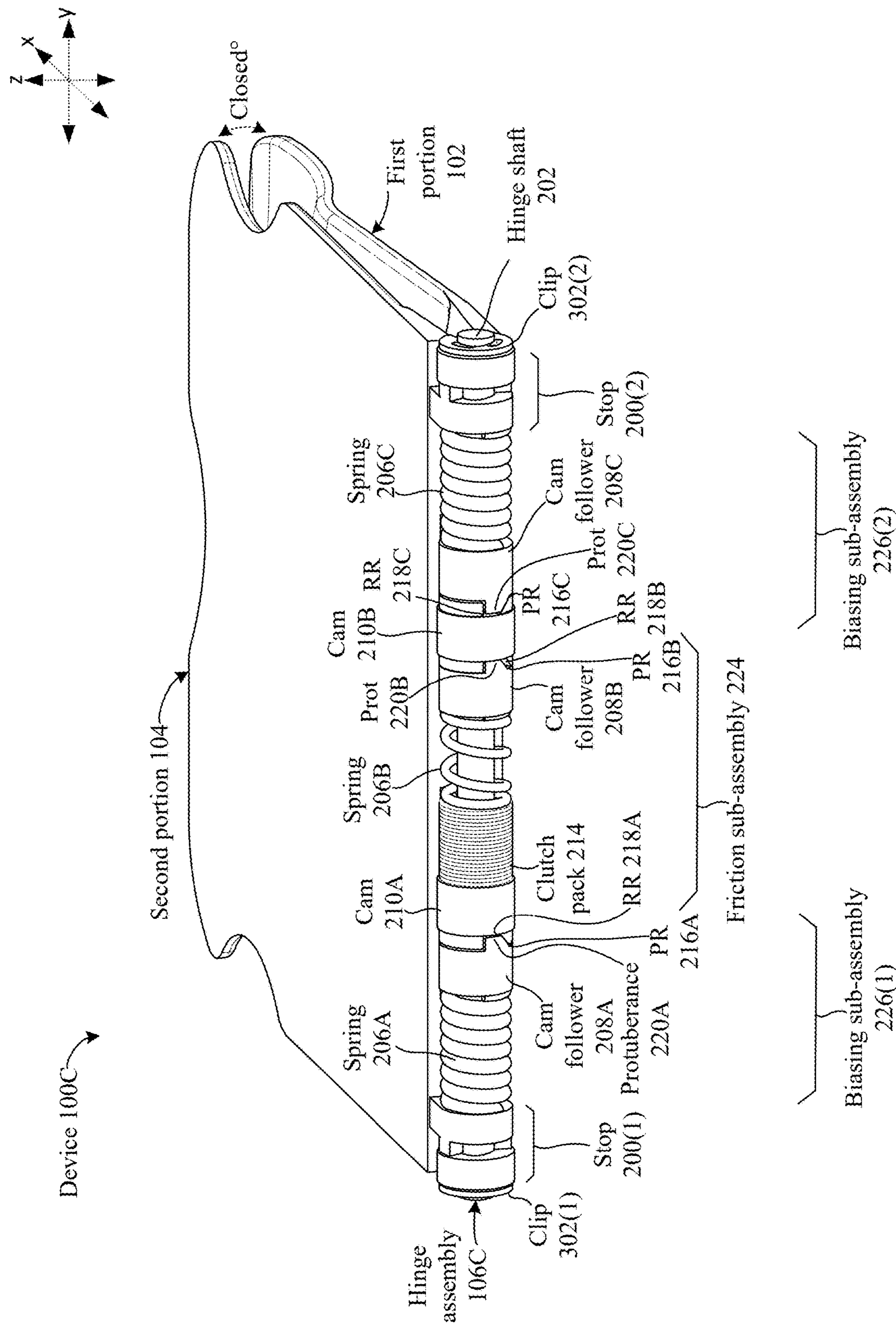


FIG. 3C

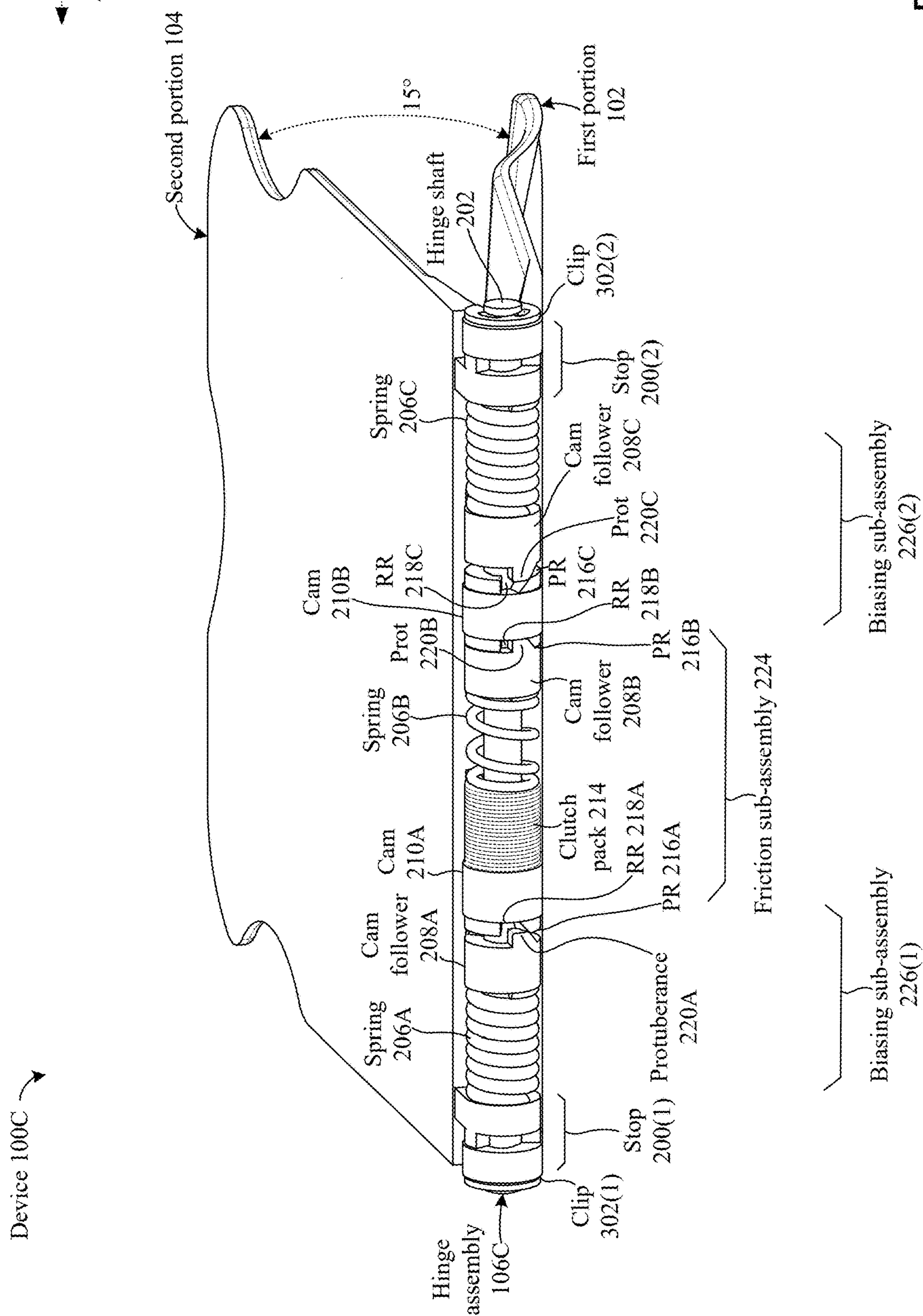
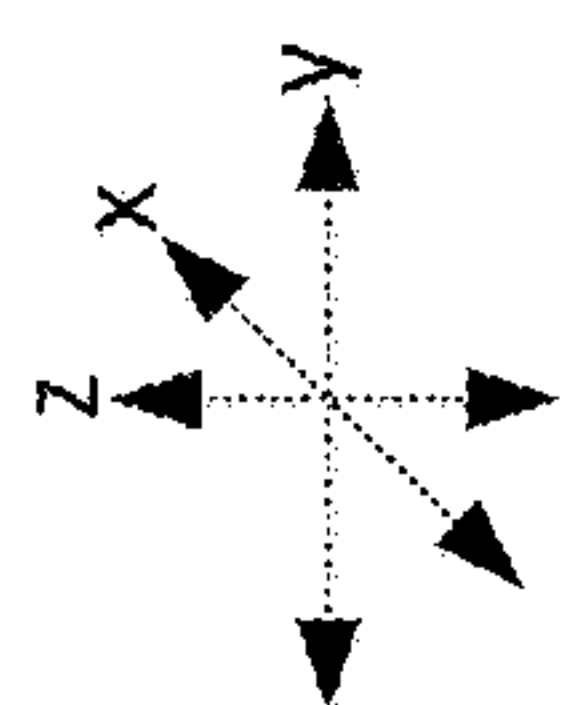
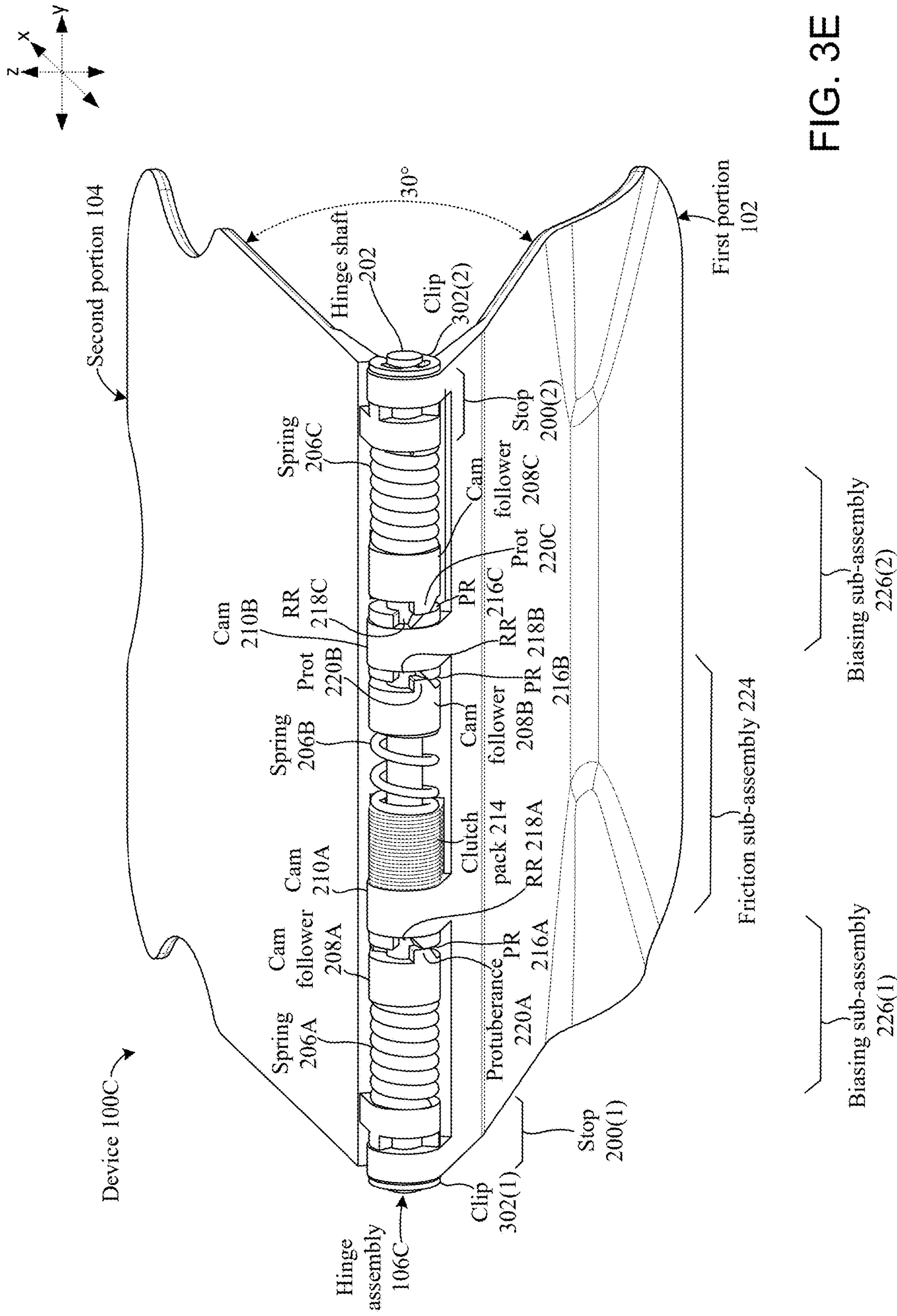


FIG. 3D



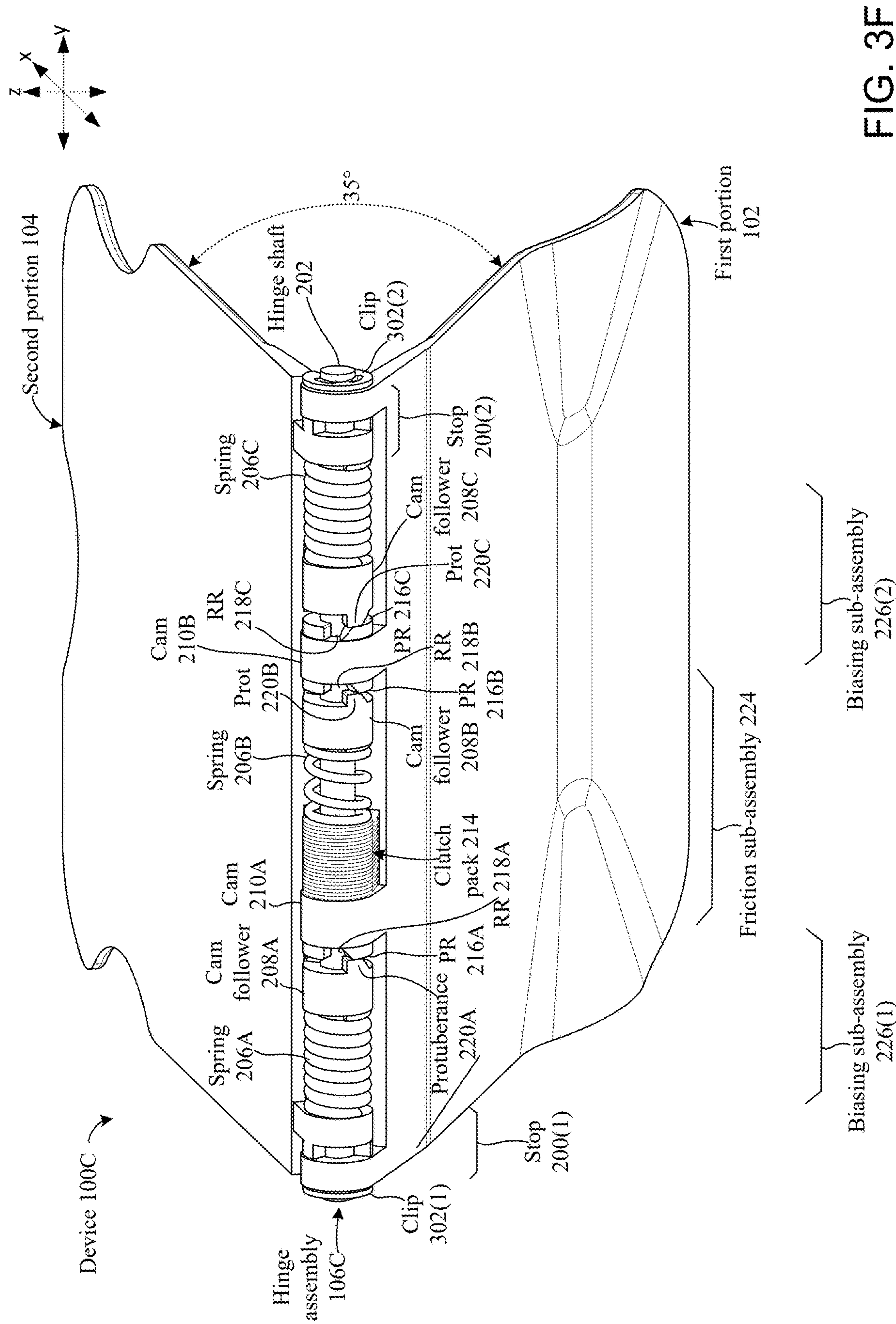


FIG. 3F

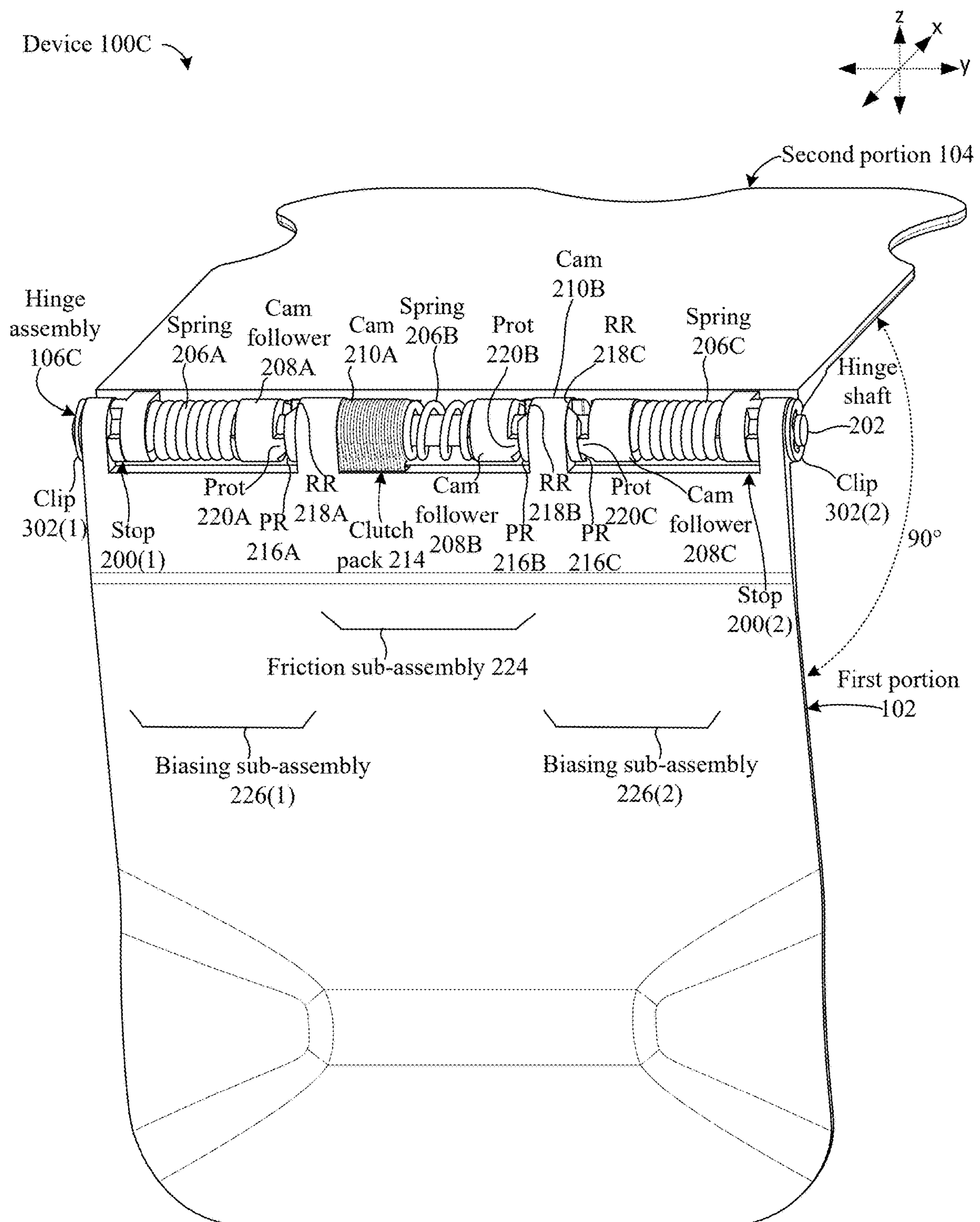


FIG. 3G

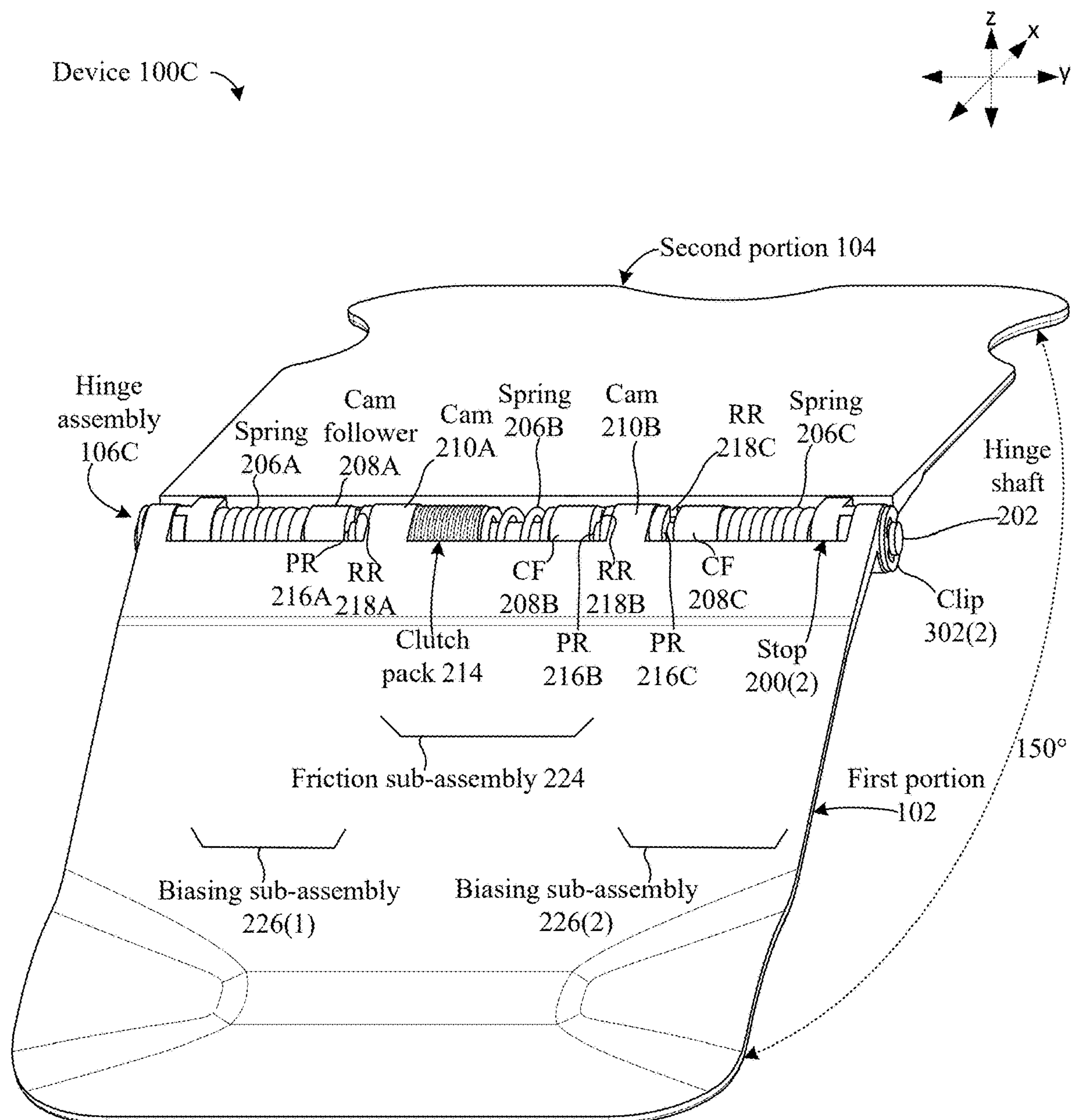


FIG. 3H

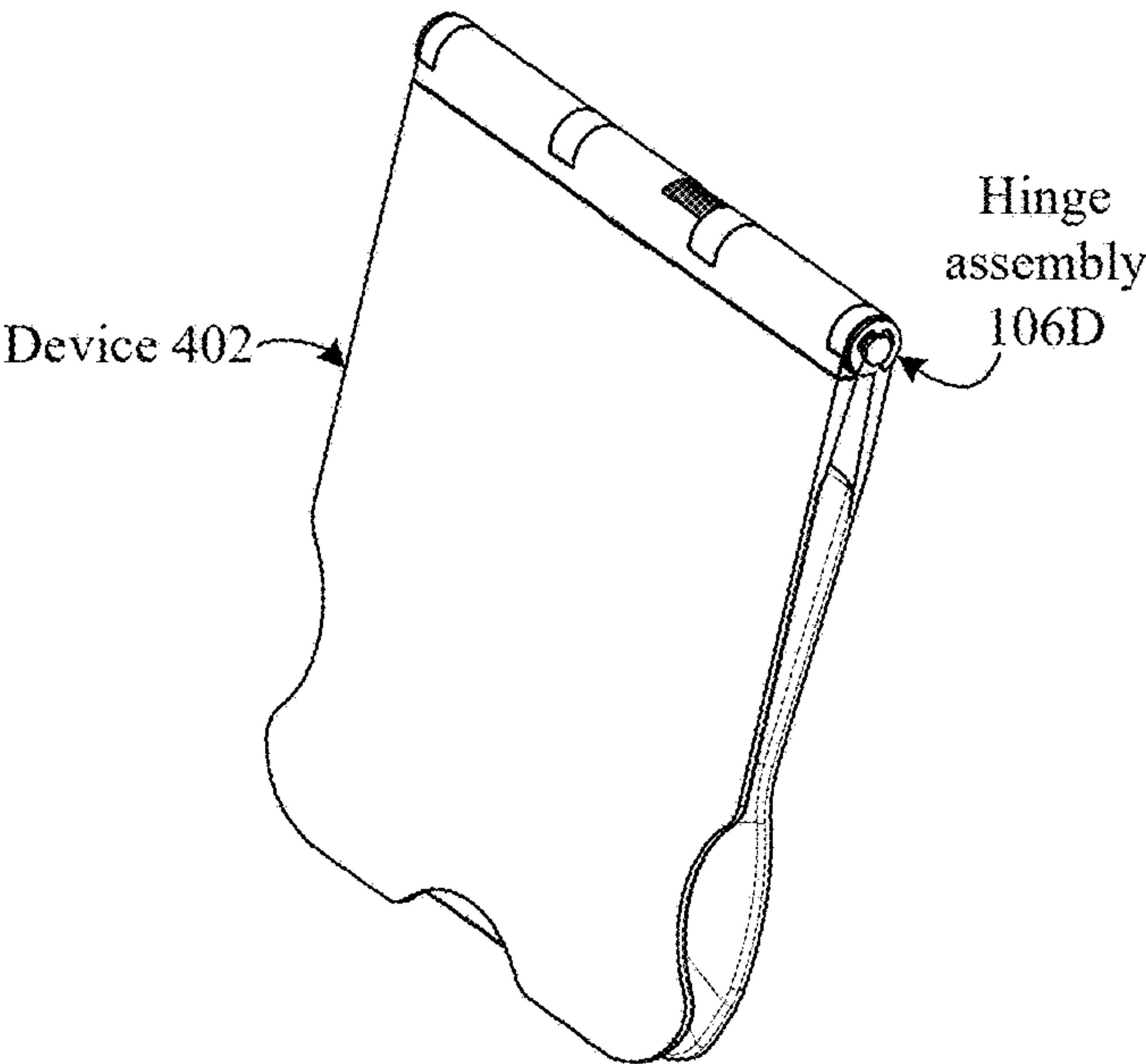


FIG. 4A

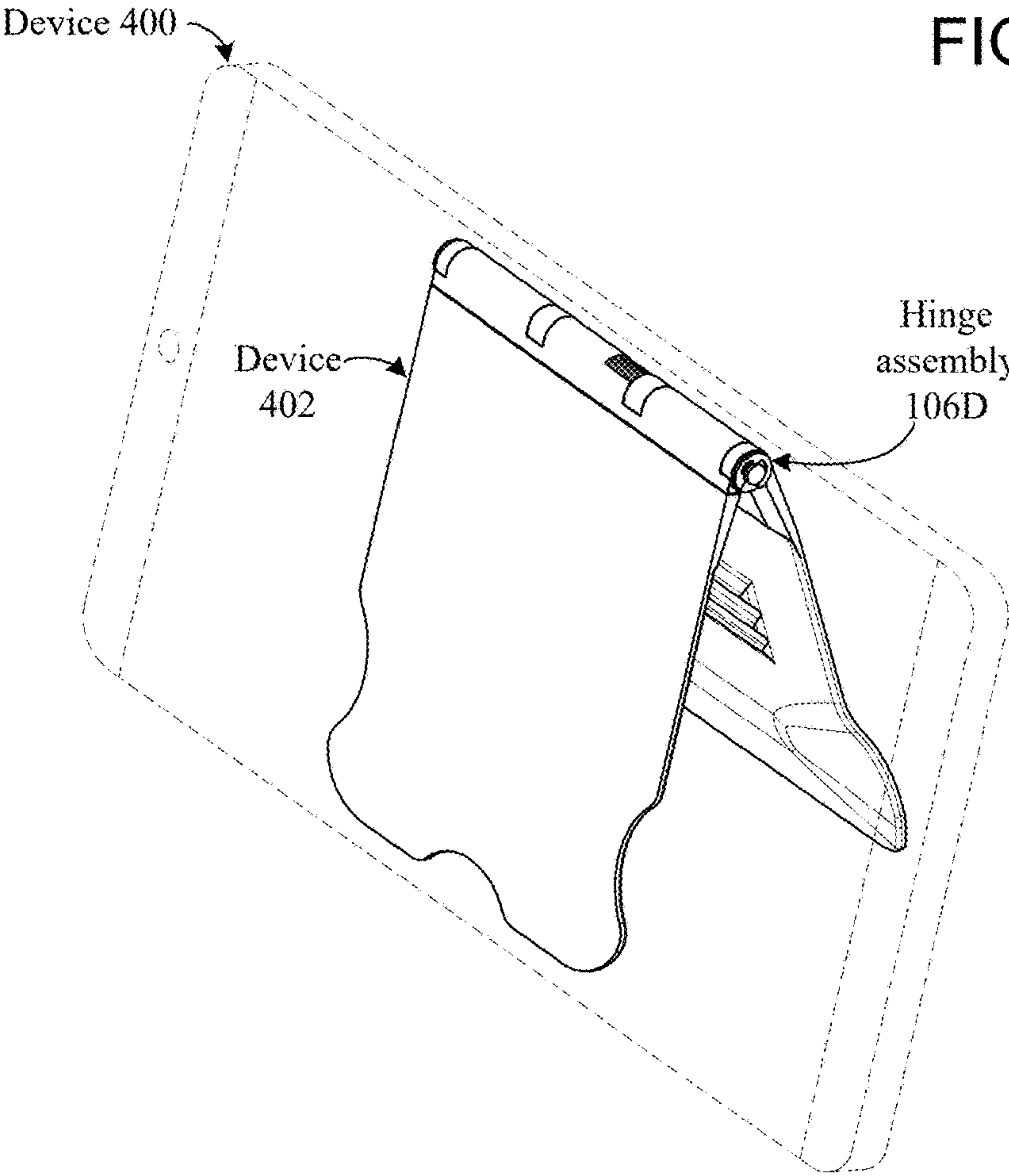


FIG. 4B

MULTI-FUNCTION HINGE

BACKGROUND

[0001] Consumers rely on many portable computer form factors in their busy lives. Some of these portable computers can include a hinge or work cooperatively with a hinge to allow the device to be orientated to different postures. For instance, smart phones, tablets, and notebook computers can provide enhanced functionality by folding for storage and opening for use. Many hinge designs have been tried for folding the device portions.

SUMMARY

[0002] This patent relates to hinged devices, such as hinged computing devices and hinges associated with these devices, such as hinges on accessory devices. The hinges can provide different functionalities at different device orientations. One example can include a first portion and a second portion that are rotatably secured by a hinge shaft through a range of rotation from a closed orientation to a fully open orientation. A friction sub-assembly can be positioned on the hinge shaft and configured to create resistance to rotation during a first sub-range of rotation that ranges from the fully opened orientation to an intermediary orientation. A biasing sub-assembly can be positioned on the hinge shaft and configured to create a bias force during a second sub-range of rotation that ranges from the intermediary orientation to the closed orientation and the biasing force biases the first and second portions to an individual orientation within the second sub-range of rotation.

[0003] This example is intended to provide a summary of some of the described concepts and is not intended to be inclusive or limiting.

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] The accompanying drawings illustrate implementations of the concepts conveyed in the present document. Features of the illustrated implementations can be more readily understood by reference to the following description taken in conjunction with the accompanying drawings. Like reference numbers in the various drawings are used wherever feasible to indicate like elements. Further, the left-most numeral of each reference number conveys the figure and associated discussion where the reference number is first introduced. Where space permits, elements and their associated reference numbers are both shown on the drawing page for the reader's convenience. Otherwise, only the reference numbers are shown. Note that some figures illustrate many elements and adding lead lines to all of the elements can diminish readability of the figure. Accordingly, not every element is designated in every figure.

[0005] FIGS. 1A-1D, 2A-2C, 3C-3H, 4A, and 4B show perspective views of example devices in accordance with some implementations of the present concepts.

[0006] FIGS. 3A and 3B show exploded perspective views of example devices in accordance with some implementations of the present concepts.

DESCRIPTION

[0007] The present concepts relate to devices, such as computing devices employing dual function hinge assemblies that can rotationally secure/couple first and second device portions. The dual function hinge assembly can

define a range of rotation for the first and second portions from fully opened to closed. The present concepts can provide different hinge functionalities depending on the orientation of the device portions. In one sub-range of orientations, such as fully opened to 30 degrees, the present concepts can provide a friction hinge functionality and at another sub-range of orientations, such as 30 degrees to closed, the present concepts can provide a biasing hinge functionality. These aspects are described in more detail below beginning with FIGS. 1A-1D. Further, the present concepts can accomplish both of these functionalities with sub-assemblies positioned on a single hinge shaft. In some cases, the friction hinge functionality can be provided by a friction sub-assembly and the biasing functionality can be provided by a biasing sub-assembly. These aspects are described in more detail below beginning with FIGS. 2A-2C.

[0008] Introductory FIGS. 1A-1D collectively show an example device **100A** that has first and second portions **102** and **104** that are rotatably secured together by a dual function hinge assembly (hereinafter, "hinge assembly") **106A** positioned in a spine **108**. The hinge assembly **106A** defines a hinge axis HA around which the first and second portions rotate. (The use of the alphabetic suffixes 'A,' 'B,' etc. relative to the device **100A** and the hinge assembly **106A** indicate that multiple different device form factors and multiple different hinge assembly form factors are described.)

[0009] The first portion **102** can extend from a hinge end **110** to a distal end **112**. The second portion **104** also can extend from a hinge end **114** to a distal end **116**. The first portion **102** can include opposing first and second major surfaces **118** and **120** (hereinafter, first and second surfaces). Similarly, the second portion **104** can include opposing first and second major surfaces **122** and **124** (hereinafter, first and second surfaces).

[0010] In some implementations, displays **126** are supported by housings **128**. For example, the displays **126** can be positioned on the first and/or second surfaces **118**, **120**, **122**, and/or **124**, respectively. In the illustrated configuration, the displays **126** are positioned on first surfaces **118** and **122**, respectively.

[0011] The hinge assembly **106A** can rotatably secure the first and second portions through a range of orientations including a closed orientation and various open orientations. The hinge assembly entails a technical solution that provides two distinct functionalities during the range of orientations. In a first sub-range of rotations, the hinge assembly provides a 'retention' or 'friction' force that acts to maintain the device portions at the orientation the user selects. In a second sub-range of rotations, the hinge assembly provides a bias force that acts to rotate the device portions to a specific orientation. In this implementation, the first sub-range of rotation is from about 180 degrees to about 30 degrees and the second sub-range of rotation is from about 30 degrees to about zero degrees. Further, in this implementation, the specific orientation is the closed orientation. Thus, from about 180 degrees to about 30 degrees, the hinge assembly provides a technical solution that holds the device at whatever specific orientation the user selects. Below 30 degrees, the hinge assembly creates a bias to close the device toward zero degrees. Stated another way, when the device is in orientations from 30 degrees to zero degrees, the device creates a bias towards the zero degree orientation. Once at

the zero-degree (e.g., closed) orientation (e.g., closed), the device continues to create the bias to maintain the closed orientation. This technical solution aids in closing the device for carrying and for creating a clamping affect that increases carrying options and decreases chances of losing the device.

[0012] FIGS. 1A and 1B show the device **100A** at representative orientations of the first sub-range of rotation. FIGS. 1C and 1D show the device at representative orientations of the second sub-range of rotation. In the first sub-range of rotation, the hinge assembly **106A** supplies the retention force that contributes to maintaining the device at the present orientation (e.g., posture). For instance, FIG. 1A shows user **130** holding the device **100A** at a 180-degree orientation (e.g., the first and second portions **102** and **104** define a 180-degree angle). The retention force maintains the device unless another force is exerted on the device, such as by the user. Thus, the hinge assembly's retention force maintains the device at the 180-degree orientation while the user holds and uses it. In this implementation, the 180-degree orientation represents the fully open orientation. However, other implementations can have a fully open orientation that is less than 180 degrees or more than 180 degrees.

[0013] For purposes of explanation, assume at this point the user wants to reduce the orientation of the device and exerts a closing force on the first and second portions. FIG. 1B shows the device rotated to the 90-degree orientation, which is still in the first sub-range of rotation. At this point, the user stops closing the device and the retention force maintains the device at the 90-degree orientation. Thus, the device maintains a 'notebook' configuration where the user can utilize the first portion **102** as an input device while viewing content on the second portion **104**. The illustrated 90-degree orientation is within the first sub-range of rotation and the retention force holds the device at the orientation. This continues as the user rotates the device portions toward one another in the closing direction through the remaining orientations of the first sub-range of rotation. Recall that in this implementation, the first sub-range of rotation ends at about a 30-degree orientation. As the device passes the 30-degree orientation, the device transitions from the first sub-range of rotation to the second sub-range of rotation. Accordingly, the device transitions from the retention force of the first sub-range of rotation to the bias force of the second sub-range of rotation. This is illustrated in FIG. 1C.

[0014] FIG. 1C shows the device continuing to rotate in the closing direction to about a 25-degree orientation due to force imparted by the user. Now, in the second sub-range of rotation (starting at 30 degrees), the biasing force creates a bias on the device portions toward a specific orientation. The bias force is represented by bias force (BF) arrow on FIG. 1C. The bias force will cause the device to rotate in the second sub-range of orientations toward the specific orientation without other external forces, such as those applied by the user. In this example, the specific orientation is the closed or zero-degree orientation, which is shown in FIG. 1D. The bias force causes the device to automatically rotate the first and second device portions toward one another in the closing direction until reaching the specific orientation or until countered by another stronger force.

[0015] FIG. 1D shows the bias force biasing the first and second portions **102** and **104** towards the closed orientation. In this case, the bias force is sufficient to clamp the device onto an object, such as the user's shirt. Here, the bias force

is large enough to keep the device from falling off as the user undertakes various activities. Thus, the bias force in the closing direction helps the user close the device and provides an attachment means so the device can be temporally attached to various objects. The present concepts provide a technical solution for a dual function hinge assembly. The hinge assembly provides the first function in the first sub-range of rotation and the second function in the second sub-range of rotation. These aspects are described in more detail below.

[0016] Note also that if the specific orientation is an orientation in the second sub-range of rotation other than the closed orientation, then the bias could be in both the closing and opening directions. For instance, if the second sub-range of rotation is closed to 30 degrees and the specific orientation is fifteen degrees, the device could provide an opening bias from closed to 15 degrees and a closing bias from 30 degrees to 15 degrees.

[0017] FIGS. 2A-2C collectively show an example hinge assembly **106B** that provides the dual functionality introduced above relative to FIGS. 1A-1D. The sequence of FIGS. 2A-2C is similar to those of FIGS. 1A-1D in that the sequence starts at the fully open orientation of FIG. 2A and rotates through a range of intermediate open orientations (represented by FIG. 2B) and finishes with the closed orientation of FIG. 2C. The hinge assembly **106B** works cooperatively with stops **200A** and **200B**. The stops **200A(1)** and **200B(1)** are associated with first portion **102** and stops **200A(2)** and **200B(2)** are associated with second portion **104**. The stops **200A** and **200B** contact one another to define the fully open orientation shown in FIG. 2A. In this implementation, the fully open orientation is about 160 degrees.

[0018] In this implementation, the hinge assembly **106B** includes hinge shaft **202**. Positioned along the hinge shaft **202** are a first compression spring (hereinafter, may be 'spring') **206A**, a first cam follower **208A**, an axial cam (hereinafter, may be 'cam') **210** that includes two cam surfaces **212**, a second cam follower **208B**, a second compression spring **206B**, and a clutch pack **214**. Each of the cam surfaces **212** includes proud regions **216** (may be abbreviated as 'PR' on drawing page for space savings) and recess regions **218** (may be abbreviated as 'RR' on drawing page for space savings). The cam followers **208** include protuberances **220** that engage the cam surfaces **212** of the axial cam **210**. (Protuberances **220** may be shortened to 'prot' on the drawing page for space savings.) Thus, a single cam **210** with opposing cam surfaces **212A** and **212B** can independently control both of the cam followers **208A** and **208B**.

[0019] The clutch pack **214** includes a set of friction plates **222**, only one of which is designated to avoid clutter on the drawing page. The compression springs **206**, axial cam **210**, cam followers **208**, friction plates **222**, and portions of the housings **128(1)** and **128(2)** are positioned on (e.g., coextensive with) the hinge shaft **202**. Stated another way, the hinge shaft **202** passes through holes defined in these elements. For instance, each friction plate **222** defines a hole through which the hinge shaft **202** passes. Alternating friction plates **222** extend to the first and second portions **102** and **104**.

[0020] The cam **210** is secured to the first portion **102** in a manner which prevents the cam **210** from moving along the hinge shaft **202** (e.g., in the y reference direction). In contrast, the cam followers **208** are secured to the second

portion **104** in a manner that allows the cam followers to move along the hinge shaft **202** (e.g., in the y reference direction). Movement of the cam followers **208** along the hinge shaft **202** (e.g., in the y reference direction) occurs when the first and second portions are rotated relative to one another (e.g., the orientation changes). Movement of cam follower **208A** in the y reference direction can affect the extent of compression of compression spring **206A**. Similarly, movement of cam follower **208B** in the y reference direction can affect the extent of compression of compression spring **206B**.

[0021] From one perspective, the cam **210** (with cam surface **212B**), cam follower **208B**, compression spring **206B**, and clutch pack **214** can function as a friction sub-assembly **224**. The friction sub-assembly **224** operates in a first sub-range of orientations to create a friction force (e.g., resistance to rotation) that operates to hold the device at its present orientation within the first sub-range of orientations. Similarly, the cam **210** (with cam surface **212A**), cam follower **208A**, and compression spring **206A** can function as a biasing sub-assembly **226**. The biasing sub-assembly **226** operates in a second sub-range of orientations to bias the first and second portions to a specific orientation with the second sub-range of orientations. Both the friction sub-assembly **224** and the biasing sub-assembly **226** are positioned on the hinge shaft **202**. Thus, the friction sub-assembly **224** and the biasing sub-assembly **226** contribute to the technical solution of the dual function hinge. Further, locating both sub-assemblies on a single hinge shaft provides a technical solution that saves space (e.g., device real estate) in the hinge assembly and decreases the number of components employed to achieve the dual functionality.

[0022] The 160-degree orientation of FIG. 2A represents an endpoint of the first sub-range of orientations. At this point, the friction sub-assembly **224** is creating a retention or friction force (e.g., resistance to rotation) that functions to hold the device at the present orientation. Specifically, protuberance **220(2)** of cam follower **208B** is contacting the proud region **216(2)** of cam surface **212B**. The contact with the proud region **216(2)** is creating a linear force on, and moving, the cam follower **208B** in the +y reference direction against the compression spring **206B**. The compression spring **206B** is captive between the cam follower **208B** and the clutch pack **214**. As such, the movement of the cam follower **208B** compresses the compression spring **206B** against the clutch pack **214** in the +y reference direction.

[0023] The clutch pack **214** is captive between the compression spring **206B** and the end of the first portion **102**. As such, the compression spring **206B** imparts a linear force in the +y reference direction on the clutch pack **214**. The linear force compresses adjacent friction plates **222** against one another. Recall that alternating friction plates **222** extend to the first and second portions **102** and **104**. The linear compressive force increases the resistance to rotation experienced by adjacent friction plates **222** and thus the first and second portions. Thus, the friction sub-assembly **224** is providing a technical solution by creating a friction force that functions to hold the device at the present orientation. Alternatively or additionally to a clutch pack, the friction sub-assembly could employ a set of detents that are configured to maintain the first and second portions at individual orientations in the first sub-range of orientations.

[0024] At this 160-degree orientation, which lies in the first sub-range of rotation, the biasing sub-assembly **226** is

not creating a biasing force. More specifically, the protuberance **220(1)** of cam follower **208A** is contacting the proud region **216(1)** of cam surface **212A**. This contact is moving the cam follower **208A** in the -y reference direction and compressing the compression spring **206A**. The proud region **216(1)** is generally flat when viewed axially along the hinge axis. Thus, the cam follower **208A** is not biased toward or away from the present orientation. Accordingly, the biasing sub-assembly **226** is not creating a biasing force on the first and second portions toward or away from the present orientation. This configuration provides a technical solution in that the biasing sub-assembly **226** is not creating a biasing force in the first sub-range of rotation that counters (e.g. acts against) the retention force generated by the friction sub-assembly **224**.

[0025] FIG. 2B shows an approximately 30-degree orientation that represents a transition from the first sub-range of rotation (e.g., 160 degrees to 30 degrees) to the second sub-range of rotation (e.g., 30 degrees to zero degrees). At this point, in relation to biasing sub-assembly **226**, the compression spring **206A** remains compressed by the cam follower **208A**. However, the protuberance **220(1)** is transitioning from the proud region **216(1)** of cam surface **212A** to the recess region **218(1)**. In the recess region **218(1)** the cam surface **212A** is sloped. The compressed compression spring **206A** continues to impart linear forces on the cam follower **208A** in the +y reference direction. These linear forces create a bias for the protuberance **220(1)** to move down the sloped surface of the recess region **218(1)**, which would allow the cam follower **208A** to move away from the compression spring **206A** and allow the compression spring **206A** to decompress.

[0026] Recall that the cam **210** is secured to the first portion **102** while the cam follower **208A** is secured to the second portion. Thus, the biasing sub-assembly **226** is beginning to create a bias on the first and second portions toward an orientation that results from the compression spring pushing the cam follower's protuberance **220(1)** into recess region **218(1)**. In this case, the protuberance **220(1)** is fully in the recess region **218(1)** at the closed orientation of FIG. 2C. Given that the cam follower **208A** is secured to the second portion **104** and the cam **210** is secured to the first portion **102**, the interaction of the protuberance **220(1)** with the recess region **218(1)** is beginning to create a bias on the first and second portions to rotate to the closed orientation.

[0027] At this point, in relation to friction sub-assembly **224** recall that during the first sub-range of orientations, the cam **210** had moved the cam follower **208B** against and compressing the compression spring **206B**. The compression spring **206B** in turn imparted a force on the friction plates **222** that increased friction (e.g., resistance to rotation) between adjacent friction plates **222**. Now, at the 30-degree orientation, the protuberance **220(2)** of cam follower **208B** of the cam surface **212B** is in the recess region **218(2)**. The compression spring **206B** has moved the cam follower **208B** in the -y reference direction and thus the compression spring **206B** can decompress.

[0028] As the compression spring **206B** decompresses, the linear force imparted by the compression spring **206B** on the clutch pack **214** is reduced and then eliminated as the spring continues to decompress. This removal of the lateral force reduces the resistance to rotation provided by the adjacent friction plates **222** of the clutch pack **214**. Thus, the clutch pack **214** no longer provides a retention force to maintain the

device (e.g., the first and second portions) at its present orientation. This provides a technical solution in that the friction sub-assembly **224** is not creating resistance to rotation that would counter (e.g., act against) the bias force created by the biasing sub-assembly **226** in the second sub-range of rotation. Thus, the biasing sub-assembly **226** is unencumbered as it biases the first and second portions toward a specific orientation. In this implementation, the specific orientation is the orientation where the protuberance **220(1)** is in the bottom of the recess region **218(1)**. In this case, the protuberance **220(1)** is in the bottom of the recess region **218(1)** in the zero degree or closed orientation as shown in FIG. 2C. Stated another way, the present concepts provide a technical solution in that the biasing sub-assembly **226** and the friction sub-assembly **224** do not counteract or ‘fight’ one another. Instead, one functions in one sub-range of rotation and the other functions in a different sub-range of rotation in a complementary manner. Without this technical solution, larger friction forces would be required in the friction sub-range and larger biasing forces would be required in the biasing sub-range. Instead, the present concepts provide a technical solution that allows reduced forces to accomplish the desired functionality and reduces stress on the device components that would occur with competing forces at a given orientation.

[0029] FIG. 2C shows the first and second portions **102** and **104** biased to the closed orientation by the biasing sub-assembly **226**. In this implementation, at this orientation, in relation to the biasing sub-assembly **226**, the protuberance **220(1)** is in the bottom of the recess region **218(1)** and is creating a bias to maintain this orientation. At this orientation (and other orientations in the second sub-range of rotation) in relation to the friction sub-assembly **224**, the protuberance **220(2)** of cam follower **208B** is fully in recess **218(2)**, which allows the cam follower **208B** to move in the $-y$ reference direction. The movement of the cam follower **208B** removes pressure on the compression spring **206B** and allows the compression spring **206B** to extend (decompress). The compression spring **206B** in turn is no longer imparting lateral forces on the clutch pack **214**. Correspondingly, the friction plates **222** are no longer forced against one another and thus do not provide resistance to rotation.

[0030] Thus, at this closed orientation of FIG. 2C, the biasing sub-assembly **226** is controlling hinge function and is biasing the first and second portions to maintain this orientation. The hinge assembly **106B** will cause the first and second portions **102** and **104** to maintain this orientation unless the user imparts a rotational force (e.g., torque) on the first and second portions sufficient to overcome the bias by causing the protuberance **220(1)** to move out of recess region **218(1)**, up the cam surface **212A** to the proud region **216(1)** while compressing compression spring **206A**.

[0031] In the second sub-range of rotation that occurs from approximately 30 degrees to zero degrees in the illustrated implementation, the effects of the friction sub-assembly **224** diminish and the effects of the biasing sub-assembly **226** increase. The effects of the friction sub-assembly **224** are nominal once the movement of the cam follower **208B** in the $-y$ reference direction allows compression spring **206B** to fully decompress. In contrast, the effects of the biasing sub-assembly **226** are significant as the protuberance **220(1)** of cam follower **208A** travels down the sloped cam surface **212A** of the recess region **218(1)**. The force of compression spring **206A** on the protuberance

220(1) creates the bias by causing the protuberance **220(1)** to continue down the sloped cam surface **212A** until the protuberance **220(1)** is bottomed out in the recess region **218(1)**. Overcoming this bias entails a user imparting enough rotational force (e.g., torque) to cause the protuberance **220(1)** to move up the sloped cam surface **212A** of the recess region **218(1)** and thereby compress the compression spring **206A**.

[0032] FIGS. 3A-3H collectively show another example device **100C**. FIGS. 3A and 3B show partially exploded views of device **100C**. FIG. 3C shows the device in a closed orientation. FIG. 3D shows the device in a 15-degree orientation. FIG. 3E shows the device in a 30-degree orientation. FIG. 3F shows the device in a 35-degree orientation. FIG. 3G shows the device in a 90-degree orientation. FIG. 3H shows the device in a fully-open 150-degree orientation. In this implementation, the closed and 30-degree orientations represent the biased sub-range and the 35, 90, and 150-degree orientations represent the friction sub-range. Note that no view shows all aspects of the device and the views should be considered collectively. Further, to provide multiple different views, FIGS. 3A and 3B show the device with the first portion **102** above the second portion **104**, whereas FIGS. 3C-3H show the device with the second portion **104** above the first portion **102**.

[0033] In this case, the hinge assembly **106C** includes three compression springs **206A**, **206B**, and **206C**, two axial cams **210A** and **210B**, and three cam followers **208A**, **208B**, and **208C**, all of which are positioned on hinge shaft **202**. The hinge assembly **106C** also includes stops **200**, clips **302**, and a cover **304**. In this implementation, the stops **200** define the range of rotation from a closed orientation of about -3.5 degrees to fully open orientation of about 150 degrees. The cam followers **208** and one side of the clutch pack **214** are received in slots in the second portion **104**. The other side of the clutch pack **214** is received in a slot in the first portion **102**. These slots are shown but not designated in FIGS. 3A and 3B. The slots for the cam followers can be slightly oversized in the y reference direction to allow movement of the cam followers **208** along the y reference axis (e.g., along the hinge axis). The clips **302** retain the other elements on the hinge shaft **202**.

[0034] The friction sub-assembly **224** entails the backside (e.g., right side on the drawing page on FIGS. 3C-3H) of cam **210A**, clutch pack **214**, compression spring **206B**, cam follower **208B**, and the left side of cam **210B**. Cam **210A** and cam **210B** are secured to the first portion in a manner that prevents them from moving along the y reference axis. The cam follower **208B** is secured to the second portion **104** and can move along the y reference axis. The clutch pack **214**, compression spring **206B**, and cam follower **208B** are captive between cam **210A** and cam **210B**. Note that the backside of cam **210A** is a flat surface rather than a cammed surface. The friction sub-assembly **224** is sandwiched (e.g., interposed) between two biasing sub-assemblies **226(1)** and **226(2)**. This configuration provides a technical solution that provides symmetrical forces along the hinge shaft **202** and thus reduces asymmetric torques that could contribute to component failure.

[0035] The biasing sub-assembly **226(1)** includes spring **206A**, cam follower **208A**, and cam **210A** in the $-y$ reference direction from the friction sub-assembly **224** (e.g., left on the drawing page of FIGS. 3C-3H). In the $+y$ reference direction from the friction sub-assembly **224** (e.g., right on

the drawing page of FIGS. 3C-3H), the biasing sub-assembly **226(2)** includes spring **206C**, cam follower **208C**, and cam **210B**. The cam followers **208A-208C** are slidably secured to the second portion **104**.

[0036] In this implementation, as shown in FIG. 3C the biasing sub-assemblies' compression springs **206A** and **206C** are preloaded (e.g., compressed at the closed orientation). Thus, at the closed orientation of FIG. 3C, compression spring **206A** is imparting a force on cam follower **208A** towards cam **210A** and similarly compression spring **206C** is imparting a force onto cam follower **208C** towards cam **210B**.

[0037] Specifically, compression spring **206A** is forcing the cam follower's protuberance **220A** into the cam's recess region **218A**. Even with the protuberance **220A** 'bottomed out' in the recess region **218A**, the preloaded compression spring **206A** is still slightly compressed and continues to exert force on the cam follower **208A** toward the cam **210A**. Similarly, compression spring **206C** is forcing the cam follower's protuberance **220C** into the cam's recess region **218C**. Even with the protuberance **220C** 'bottomed out' in the recess region **218C** the preloaded compression spring **206C** is still slightly compressed and continues to exert force on the cam follower **208C** toward the cam **210B**. As will be described in more below, rotating the first and second portions **102** and **104** will entail imparting force on the first and second portions to further compress compression springs **206A** and **206C**. Thus, the present concepts provide a technical solution where the device is biased to maintain the closed orientation unless sufficient force is imparted on the device to overcome the bias.

[0038] The biasing sub-assemblies **226** provide a closed orientation to a 30-degree orientation axial cam functionality. Specifically, cam follower **208A** and axial cam **210A** operate to provide a closed orientation to a 30-degree orientation axial cam functionality. Similarly, cam follower **208C** and cam **210B** operate to provide a closed to a 30-degree axial cam functionality. In this example, the biasing sub-assemblies **226** bias (e.g., closing biasing torque) the first and second portions to (e.g., toward) the closed orientation. Alternatively, the biasing sub-assemblies **226** could bias (e.g., opening biasing torque) the first and second portions to (e.g., away from) the closed orientation. In this case, the bias to the closed orientation is achieved because the end of protuberance **220A** is approximately as wide (angular span relative to the hinge axis) as the bottom of the recess region **218A**. Similarly, the end of protuberance **220C** is approximately as wide (angular span relative to the hinge axis) as the bottom of the recess region **218C**. Thus, in the closed orientation, the end of protuberances **220A** and **220C** are bottomed out in recess regions **218A** and **218C**, respectively with little or no free rotation.

[0039] As shown in FIG. 3D, for rotation to occur from the closed orientation, the protuberances **220A** and **220C** have to climb the slanted cam surfaces **212A** and **212C**, respectively. Climbing the slanted cam surfaces **212A** and **212C** compresses springs **206A** and **206C**. Thus, sufficient rotational force has to be imparted on the first and second portions from an external source to overcome the biases imparted by the compression springs **206A** and **206C**. The protuberances **220A** and **220C** will reach the top of the slanted cam surfaces **212A** and **212C** and transition to the proud regions **216A** and **216C** at about the 30-degree

orientation as shown in FIG. 3E. Transitioning to the proud regions **216A** and **216C** ends the bias sub-rotation.

[0040] In contrast, the friction sub-assembly's compression spring **206B** is not preloaded (e.g., is fully extended in the closed orientation of FIG. 3C). Alternatively, spring **206B** is preloaded with a small amount of force to prevent rattle within the clutch pack **214**. Cam **210B** and cam follower **208B** operate to provide a 15-degree to 35-degree axial cam functionality. The recess region **218B** has a greater angular span, by about 15 degrees in this example, than the end of protuberance **220B**. Thus, as shown in FIGS. 3C and 3D, there is 15 degrees of free rotation from the closed orientation before the end of protuberance **220B** contacts the slanted cam surfaces **212B**. At that orientation, the end of protuberance **220B** contacts the slanted cam surfaces **212B**. As shown in FIG. 3E, further rotation causes the end of protuberance **220B** to climb the slanted cam surfaces **212B**, which moves the cam follower **208B** in the -y reference direction. As the protuberance **220B** rides up the cam surface **212B**, the cam follower **208B** is forced to move linearly and begins to compress the spring **206B** and affects the clutch pack **214**. The effect on the clutch pack **214** peaks at the proud region (e.g., top (flat)) **216B** of the cam surface **212B** as shown in FIG. 3F.

[0041] As shown in FIG. 3F, which is about 35 degrees of rotation, the interaction of the end of protuberance **220B** and the slanted cam surfaces **212B** compresses the spring **206B** and is imparting a linear force on the clutch pack **214**. The linear force on the clutch pack **214** creates a resistance to rotation of the first and second portions and starts the friction sub-rotation. The friction sub-rotation is maintained while the protuberance **220B** remains on the proud region **216B** from this point to the fully open orientation. Note, cams **210A** and/or **210B** and their associated protuberances **220** can be adjusted in slop angle, shape, and/or position, to tune the desired range of torques or friction throughout the system.

[0042] FIG. 3G shows an intermediate orientation of the friction sub-rotation of about 90 degrees. During the friction sub-rotation, which in this implementation is from the 35-degree orientation to the fully open 150-degree orientation, the friction sub-assembly **224** provides friction between the adjacent friction plates of the clutch pack **214**. In contrast, the biasing sub-assemblies **226** are not creating a bias because the protuberances **220A** and **220C** continue to contact the proud regions **216A** and **216C** of cams **210A** and **210B**, respectively. The proud regions **216A** and **216C** are flat and thus the interaction with the protuberances does not create a bias to any orientations in the friction sub-rotation.

[0043] FIG. 3H shows the first and second portions **102** and **104** in the fully open orientation of about 150 degrees. Contact between protuberance **220B** and proud region **216B** continues to push the cam follower **208B** in the -y reference direction (e.g., away from cam **210B**). In turn, the cam follower **208B** compresses the compression spring **206B** against the clutch pack **214**. The clutch pack's friction plates are forced together by the compressed spring **206B**. The spring **206B** remains compressed in the friction sub-rotation because the protuberance **220B** of the cam follower **208B** remains on the proud region **216B** of axial cam **210B**. The friction from the clutch pack **214** provides a technical solution of resistance to rotation to hold the first and second portions **102** and **104** at an individual orientation unless acted upon by an external force.

[0044] At this point, and throughout the friction sub-range of rotation, the biasing sub-assemblies 226 are not creating a bias to an individual orientation because the protuberances 220A and 220C, which are not readily visible at this orientation, continue to contact the proud regions 216A and 216C of cams 210A and 210C, respectively. The proud regions 216A and 216C are flat and thus the interaction with the protuberances 220A and 220C does not create a bias to any orientations in the friction sub-rotation. Thus, the present concepts provide a technical solution where the hinge assembly offers two different hinge functionalities depending on the orientation. Further, the functionalities are range specific (e.g., limited to sub-ranges of rotation) and thus do not counter or inhibit each other.

[0045] Various dual function hinge assemblies are described above that have a friction function in a first sub-range of rotation and a biasing function in a second sub-range of rotation. FIGS. 1A-1D show an example dual function hinge assembly integral in a folding computing device. FIGS. 2A-3H show details of example mechanisms of some hinge assembly implementations.

[0046] FIGS. 4A and 4B collectively show another dual function hinge assembly 106D that can be associated with a device 400. In this case, the hinge assembly 106D is integrated into an accessory device 402. FIG. 4A shows the accessory device 402 in isolation. The accessory device 402 is in the first (e.g., biased) sub-range of rotation and is providing the bias functionality.

[0047] FIG. 4B shows the accessory device 402 secured to device 400. The accessory device 402 is now in the second (e.g., friction) sub-range of rotation. The accessory device 402 is providing a friction functionality and is holding the device 400 in a desired posture. In various implementations, the accessory device 402 can be manifest as a case that snaps onto and protects the device 400. Alternatively, the accessory device 402 can be manifest as a clip that is removably secured to the device 400, such as by magnets. Other form factors are contemplated.

[0048] Individual elements of the hinge assemblies can be made from various materials, such as metals, plastics, and/or composites. These materials can be prepared in various ways, such as in the form of sheet metals, die cast metals, machined metals, 3D printed materials, molded or 3D printed plastics, and/or molded or 3D printed composites, among others, and/or any combination of these materials and/or preparations can be employed.

[0049] The present hinge assembly concepts can be utilized with any type of device, such as but not limited to notebook computers, smart phones, wearable smart devices, tablets, and/or other types of existing, developing, and/or yet to be developed devices.

[0050] Various methods of manufacture, assembly, and/or use for hinge assemblies and devices are contemplated beyond those shown above relative to FIGS. 1A-4B.

[0051] Although techniques, methods, devices, systems, etc., pertaining to hinge assemblies are described in language specific to structural features and/or methodological acts, it is to be understood that the subject matter defined in the appended claims is not limited to the specific features or acts described. Rather, the specific features and acts are disclosed as example forms of implementing the claimed methods, devices, systems, etc.

[0052] Various examples are described above. Additional examples are described below. One example includes a

device comprising a first portion and a second portion that are rotatably secured by a hinge shaft through a range of rotation from a closed orientation to a fully open orientation, a friction sub-assembly positioned on the hinge shaft and configured to create resistance to rotation during a first sub-range of rotation from the fully opened orientation to an intermediary orientation, and a biasing sub-assembly positioned on the hinge shaft and configured to create a bias force towards the closed orientation during a second sub-range of rotation from the intermediary orientation to the closed orientation.

[0053] Another example can include any of the above and/or below examples where the friction sub-assembly comprises a clutch pack positioned on the hinge shaft and a first compression spring positioned on the hinge shaft, and wherein the biasing sub-assembly comprises a second compression spring positioned on the hinge shaft.

[0054] Another example can include any of the above and/or below examples where the device further comprises an axial cam positioned on the hinge shaft and configured to cause the first compression spring to be relatively more compressed in the first sub-range of rotation and relatively less compressed in the second sub-range of rotation.

[0055] Another example can include any of the above and/or below examples where the axial cam is positioned between the friction sub-assembly and the biasing sub-assembly.

[0056] Another example can include any of the above and/or below examples where the axial cam comprises a single axial cam that acts on both the friction sub-assembly and the biasing sub-assembly.

[0057] Another example can include any of the above and/or below examples where the axial cam comprises a first axial cam that is configured to act on the friction sub-assembly and a second axial cam that is configured to act on the biasing sub-assembly.

[0058] Another example can include any of the above and/or below examples where in the first sub-range of rotation the relatively more compressed first compression spring is configured to impart a force on the clutch pack and in the second sub-range of rotation the relatively less compressed first compression spring is configured not to impart a force on the clutch pack.

[0059] Another example can include any of the above and/or below examples where the device further comprises a second biasing sub-assembly positioned along the hinge shaft.

[0060] Another example can include any of the above and/or below examples where the friction sub-assembly is interposed along the hinge shaft between the biasing sub-assembly and the second biasing sub-assembly.

[0061] Another example can include any of the above and/or below examples where the device further comprises a first axial cam positioned on the hinge shaft between the biasing sub-assembly and the friction sub-assembly and a second axial cam positioned on the hinge shaft between the friction sub-assembly and the second biasing sub-assembly.

[0062] Another example includes a device comprising a first portion that includes a first display and a second portion that includes a second display, the first and second portions are rotatably secured by a hinge shaft through a range of rotation from a closed orientation to a fully open orientation, a friction sub-assembly positioned on the hinge shaft and configured to create resistance to rotation during a first

sub-range of rotation that ranges from the fully opened orientation to an intermediary orientation, and a biasing sub-assembly positioned on the hinge shaft and configured to create a bias force during a second sub-range of rotation that ranges from the intermediary orientation to the closed orientation and the biasing force biases the first and second portions to an individual orientation within the second sub-range of rotation.

[0063] Another example can include any of the above and/or below examples where the biasing force is a closing biasing torque and the individual orientation is the closed orientation.

[0064] Another example can include any of the above and/or below examples where the biasing force is an opening biasing torque and the individual orientation is the intermediary orientation.

[0065] Another example can include any of the above and/or below examples where the intermediary orientation lies within a range of 10 degrees to 90 degrees.

[0066] Another example can include any of the above and/or below examples where the intermediary orientation is 30 degrees.

[0067] Another example can include any of the above and/or below examples where the biasing sub-assembly comprises a clutch pack.

[0068] Another example can include any of the above and/or below examples where the biasing sub-assembly comprises multiple detents associated with orientations within the first sub-range of rotation.

[0069] Another example can include any of the above and/or below examples where the device comprises a computing device or wherein the device comprises an accessory for a computing device.

[0070] Another example includes a device comprising a first portion and a second portion that are rotatably secured by a hinge shaft through a range of rotation from a closed orientation to a fully open orientation, a friction sub-assembly positioned on the hinge shaft and configured to create resistance to rotation during a first sub-range of rotation between the fully opened orientation and an intermediary orientation and to not create resistance to rotation during a second sub-range of rotation between the intermediary orientation and the closed orientation, and a biasing sub-assembly positioned on the hinge shaft and configured to create a bias force that biases the first and second portions to an individual orientation within the second sub-range of rotation and to not create a bias force that biases the first and second portions during the first sub-range of rotation.

[0071] Another example can include any of the above and/or below examples where the device further comprises an axial cam positioned on the hinge shaft and configured to rotate with rotation of the first and second portions and wherein the axial cam is configured to activate the friction sub-assembly and not the biasing sub-assembly in the first sub-range of rotation and to activate the biasing sub-assembly and not the friction sub-assembly in the second sub-range of rotation.

1. A device, comprising:

- a first portion and a second portion that are rotatably secured by a hinge shaft through a range of rotation from a closed orientation to a fully open orientation;
- a friction sub-assembly positioned on the hinge shaft and configured to create resistance to rotation during a first

sub-range of rotation from the fully opened orientation to an intermediary orientation; and,

a biasing sub-assembly positioned on the hinge shaft and configured to create a bias force towards the closed orientation during a second sub-range of rotation from the intermediary orientation to the closed orientation.

2. The device of claim 1, wherein the friction sub-assembly comprises a clutch pack positioned on the hinge shaft and a first compression spring positioned on the hinge shaft, and wherein the biasing sub-assembly comprises a second compression spring positioned on the hinge shaft.

3. The device of claim 2, further comprising an axial cam positioned on the hinge shaft and configured to cause the first compression spring to be relatively more compressed in the first sub-range of rotation and relatively less compressed in the second sub-range of rotation.

4. The device of claim 3, wherein the axial cam is positioned between the friction sub-assembly and the biasing sub-assembly.

5. The device of claim 3, wherein the axial cam comprises a single axial cam that acts on both the friction sub-assembly and the biasing sub-assembly.

6. The device of claim 3, wherein the axial cam comprises a first axial cam that is configured to act on the friction sub-assembly and a second axial cam that is configured to act on the biasing sub-assembly.

7. The device of claim 3, wherein in the first sub-range of rotation the relatively more compressed first compression spring is configured to impart a force on the clutch pack and in the second sub-range of rotation the relatively less compressed first compression spring is configured not to impart a force on the clutch pack.

8. The device of claim 1, further comprising a second biasing sub-assembly positioned along the hinge shaft.

9. The device of claim 8, wherein the friction sub-assembly is interposed along the hinge shaft between the biasing sub-assembly and the second biasing sub-assembly.

10. The device of claim 9, further comprising a first axial cam positioned on the hinge shaft between the biasing sub-assembly and the friction sub-assembly and a second axial cam positioned on the hinge shaft between the friction sub-assembly and the second biasing sub-assembly.

11. A device, comprising:

a first portion that includes a first display and a second portion that includes a second display, the first and second portions are rotatably secured by a hinge shaft through a range of rotation from a closed orientation to a fully open orientation;

a friction sub-assembly positioned on the hinge shaft and configured to create resistance to rotation during a first sub-range of rotation that ranges from the fully opened orientation to an intermediary orientation; and,

a biasing sub-assembly positioned on the hinge shaft and configured to create a bias force during a second sub-range of rotation that ranges from the intermediary orientation to the closed orientation and the biasing force biases the first and second portions to an individual orientation within the second sub-range of rotation.

12. The device of claim 11, wherein the biasing force is a closing biasing torque and the individual orientation is the closed orientation.

13. The device of claim **11**, wherein the biasing force is an opening biasing torque and the individual orientation is the intermediary orientation.

14. The device of claim **11**, wherein the intermediary orientation lies within a range of 10 degrees to 90 degrees.

15. The device of claim **14**, wherein the intermediary orientation is 30 degrees.

16. The device of claim **11**, wherein the biasing sub-assembly comprises a clutch pack.

17. The device of claim **16**, wherein the biasing sub-assembly comprises multiple detents associated with orientations within the first sub-range of rotation.

18. The device of claim **11**, wherein the device comprises a computing device or wherein the device comprises an accessory for a computing device.

19. A device, comprising:

a first portion and a second portion that are rotatably secured by a hinge shaft through a range of rotation from a closed orientation to a fully open orientation;

a friction sub-assembly positioned on the hinge shaft and configured to create resistance to rotation during a first

sub-range of rotation between the fully opened orientation and an intermediary orientation and to not create resistance to rotation during a second sub-range of rotation between the intermediary orientation and the closed orientation; and,

a biasing sub-assembly positioned on the hinge shaft and configured to create a bias force that biases the first and second portions to an individual orientation within the second sub-range of rotation and to not create a bias force that biases the first and second portions during the first sub-range of rotation.

20. The device of claim **19**, further comprising an axial cam positioned on the hinge shaft and configured to rotate with rotation of the first and second portions and wherein the axial cam is configured to activate the friction sub-assembly and not the biasing sub-assembly in the first sub-range of rotation and to activate the biasing sub-assembly and not the friction sub-assembly in the second sub-range of rotation.

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